Positive-Data Learning of Multiple Context-Free Languages under Fixed Finite-Monoid Typing

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Abstract

Fix a fan-out bound f and a homomorphism $h : \Sigma^* \to M$ into a finite monoid. We study positive-data learning of multiple context-free languages when tuples may be interchanged only if corresponding components have equal h-values. At bounded rule rank this class is an instance of Yoshinaka's general substitutable-signature framework; we verify the relationship for the guarded observed-tuple construction, and relate its uncapped extension to Yoshinaka's uncapped learner.

Our principal quantitative question is the *total length* of characteristic positive data, measured in the size and rule thickness of an original MCFG presentation. For typings determined by a fixed prefix–suffix window, the total length is polynomial in both parameters at every fixed fan-out and rule rank: the proof preserves boundary information across several tuple components simultaneously. For a fixed prefix–suffix window, the characteristic-data total length is $O(\|G\|^2\bar{\theta}_G)$. Combining such a window with an arbitrary fixed finite-group typing gives the separate bound $O(\|G\|^3\bar{\theta}_G)$; an Erdős–Szekeres-type block-order shortening argument removes commutativity. For abelian groups a sharper explicit constant follows from zero sums. In particular, finite locally trivial positive-image typings have the quadratic bound, while typings whose positive image belongs to their pseudovariety join with finite groups have the cubic bound. At rule rank one, every fixed typing admits characteristic data polynomial in grammar size alone. For arbitrary finite typings, a version-locked, publicly accessible companion CFG manuscript gives a counterexample to universal original-thickness polynomial bounds; the window and window–finite-group conditions avoid that obstruction. We also establish finite-typing, regular-filter and fan-out separations, including a boundary-typed non-context-free target of minimum fan-out and minimum rule rank two.

Keywords: grammatical inference; multiple context-free grammars; positive data; finite-monoid typing; multidimensional substitutability; Gold identification; rule rank.

1 Introduction

Positive-data grammatical inference asks when a language can be recovered from an enumeration of its positive examples alone [1,2]. Distributional approaches make this possible for restricted grammar classes by turning observed interchangeability of substrings or tuples into grammar nonterminals. For context-free grammars (CFGs) this includes substitutable languages [3] and Yoshinaka's (k, ℓ) hierarchy [12]. Yoshinaka extended the same idea to multiple context-free grammars (MCFGs), where nonterminals generate tuples and substitution is tested by multi-hole sentence contexts [13,14]. MCFGs form a standard mildly context-sensitive formalism and can directly represent discontinuous and cross-serial dependencies [30,31].

The general learning principle and its scope. Yoshinaka's 2011 observed-tuple learner gives the untyped MCFG baseline [14]. His later signature framework includes bounded-arity MCFGs, finite refinements of sorts and prefix–suffix typing; it already provides positive-data identification, polynomial-time updates and characteristic sets of bounded cardinality [15, Theorem 1, Lemma 3 and Examples 3–4]. For bounded rank, Appendix B proves an exact signature characterization of our typed class and a correspondence of reconstruction operators. Thus the qualitative learning principle is prior work. The questions addressed here are the *total length* of characteristic positive data relative to an original MCFG presentation and separations involving finite typing, fan-out and rule rank. A recent *substitution lemma* for MCFL nonmembership [23] addresses a different question: it is a necessary condition for MCFL membership, not a distributional positive-data learning criterion.

This paper fixes, as part of the learnable class, a homomorphism $h : \Sigma^* \to M$ into a finite monoid and compares observed tuples only when corresponding components have equal h-values. The compositional type guard is fixed before the positive text is seen, separating the choice of a finite bias from learning under that bias [5]. Prefix–suffix and DFA transition morphisms provide two useful families. A fixed prefix–suffix window is a uniform, independently chosen finite parameter of the target class, not a parameter tuned to each sample; the principal total-length bound applies to every target in that fixed window-typed class at the given fan-out and rule rank. Fan-out is part of the same bias because it determines which tuple arities are tested for substitution. Thus a fixed typing supplies finite-state information while the MCFG carries the unbounded dependency.

Other learning biases make different choices. Takada uses regular control languages over derivations [6, 7], while local substitutability restricts the evidence used to compare factors [9, 11]. Positive-data learning has also been studied for other mildly context-sensitive formalisms [19]. MCFG learning with richer information includes membership-query and dual context-set approaches [16, 18, 20]. The present paper instead remains positive-data-only, uses observed tuple values as the basic sample-derived objects, and lets a fixed finite typing guard substitutions.

Contributions. Our results focus on the total *length* of characteristic positive data, rather than the known existence or cardinality of finite characteristic sets.

(1) *A multi-component quantitative upper bound.* At every fixed fan-out and rule rank, a typing determined by a fixed prefix–suffix window admits characteristic data of total length $O(\|G\|^2 \max\{1, \theta_G\})$ for a reduced good presentation G (Theorem 4.9). Its key ingredient is a short-tuple lemma preserving boundary information across all tuple components simultaneously. Appendices C and D extend the result to a window combined with any fixed finite group (including nonabelian groups), with total length $O(\|G\|^3 \bar{\theta}_G)$ (Theorem D.7). The abelian case admits the smaller explicit constant of Theorem C.4. An algebraic criterion (Corollary 4.11) reformulates these guarantees for positive-image semigroups in **LI** and **LI** $\vee$ **Grp**.

(2) *Typing, fan-out and rule-rank separations.* We give languages outside every regular-filter fragment of Yoshinaka's untyped class (Theorem 5.5), a fan-out-one target that fails the fan-out-two substitution condition under *every* finite typing (Proposition 5.3), and a boundary-typed non-context-free witness with minimum fan-out and rule rank two (Theorem 5.2).

The learning operator itself is a componentwise-typed instance of Yoshinaka's observed-tuple construction; its bounded-rank placement within the general signature framework is given by Proposition B.2.

Relation to prior work and scope. The observed-tuple grammar, conservative update policy and uncapped qualitative identification come from Yoshinaka [13, 14]; the general finite-sort framework and its characteristic-cardinality bound are due to Yoshinaka [15]. Our direct soundness/completeness proof verifies the particular componentwise guarded constructor; it does not claim a new general learning mechanism. The rank-one characteristic-length result is an elementary refinement consequence; the substantial higher-rank statement is the bound obtained from simultaneous preservation of multiple tuple-component boundaries (Theorem 4.9).

At fan-out one, polynomial characteristic data for fixed prefix–suffix typing were obtained by Yoshinaka [12, Theorem 1]. The version-locked companion CFG manuscript [25] develops arbitrary finite typing and establishes the operator-independent fan-out-one lower bound. These are prior results, not standalone new MCFG contributions here. The MCFG-specific questions are componentwise typing of tuples, characteristic-data length at higher rank, and interactions with fan-out and regular filtering. The non-branching lower bound used for the intrinsic-rank witness is prior work; the contribution is its boundary-typed wrapper, typed membership and Yoshinaka separation, and its connection to the quantitative theorem.

Throughout the polynomial statements, fan-out, rank cap, and finite typing are fixed. Characteristic data for the set-driven reconstruction concern language-level exactness; syntactic stabilization is supplied by the conservative sequential learner. The full-refinement and empty-component obstructions concern only those constructions, not all learners. The distinct operator-independent lower bound at fan-out one is due to the companion CFG paper [25].

Sections 2–4 develop the model, reconstruction, convergence, and quantitative bounds; Section 5 gives the comparison with Yoshinaka's classes and the separation results.

Proof methodology and AI-assisted research. This is a theoretical study: the claims are justified by the formal proofs presented here rather than by computational experiments or automated reviews. Generative AI tools were used to explore and critique candidate arguments and to assist manuscript revisions; such output is not itself mathematical verification. The tools and their roles are disclosed at the end of the manuscript.

2 Preliminaries

Throughout, Σ denotes a finite terminal alphabet, ε denotes the empty word, Σ^* is the set of all finite words over Σ, and $\Sigma^+ := \Sigma^* \setminus \{\varepsilon\}$.

2.1 Positive-Data Learning

We use Gold's explanatory learning model and restate the needed conventions so that the MCFG development is self-contained.

Definition 2.1 (Text and TxtEx identification)**.** We use Gold's explanatory identification criterion [1]. To handle the empty language, fix a pause symbol $\perp \notin \Sigma$; thus $\perp, \perp, \ldots$ is a text for the empty language. A *text* for $L \subseteq \Sigma^*$ is an infinite sequence over $L \cup \{\perp\}$ in which every element of L appears at least once. Pause symbols may occur arbitrarily, and $\perp$ is not an end-of-text marker.

The *content* of a finite prefix is the finite set $K \subseteq L$ of positive examples from Σ^* that occur in that prefix. Thus neither the positions nor the number of pause symbols, nor repeated occurrences of the same positive example, affect the content. A learner is a computable function mapping each finite prefix to a hypothesis grammar.

A learner *TxtEx-identifies* L from positive data if, for every text for L, there are n_0 and a single hypothesis grammar H such that all stages from n_0 onward output exactly H, with $L(H) = L$. It TxtEx-identifies a class when it does so for every language in the class. A sequential learner is *conservative* if it keeps its current hypothesis whenever the newly observed positive datum is already in that hypothesis language, changing it only when the datum is not generated.

For later use, fix a distinguished hypothesis code H_0 with $L(H_0) = \emptyset$.

Definition 2.2 (Set-driven reconstruction operator and characteristic sample)**.** A *set-driven reconstruction operator* $\mathcal{B}$ maps each finite positive sample K to a grammar $\mathcal{B}(K)$, depending only on the set K, not on presentation order or multiplicity. It is *sample-consistent* if $K \subseteq L(\mathcal{B}(K))$ for every finite K.

Following the characteristic-set viewpoint of de la Higuera [4], for a target language L, a finite set $K^\star \subseteq L$ is a *characteristic sample for* $\mathcal{B}$ if, for every finite set K, $K^\star \subseteq K \subseteq L$ implies $L(\mathcal{B}(K)) = L$. The case $K^\star = \emptyset$ is allowed. We use *characteristic data* as a neutral shorthand for the positive content of such a sample; whenever a polynomial bound is claimed, we state explicitly whether it concerns cardinality or total positive length.

This finite-sample notion is distinct from TxtEx learning: $\mathcal{B}(K)$ need not stabilize syntactically as the accumulated sample grows. The sequential learner will be introduced separately after exact reconstruction is proved.

Definition 2.3 (Positive sample size)**.** For finite $K \subseteq \Sigma^*$, define $\|K\|_+ := \sum_{w \in K} (|w| + 1)$. For a tuple $\vec{u} = (u_1, \ldots, u_d)$, write $\|\vec{u}\|_1 := \sum_{i=1}^{d} |u_i|$ for its total component length.

2.2 Finite-Monoid Typing

Definition 2.4 (Finite typing representation)**.** A finite-monoid homomorphism $h\colon \Sigma^* \to M$ is represented by the multiplication table of M together with the letter images $h(a)$, $a \in \Sigma$. Hence $h(w)$ is computable by a left-to-right product in linear time in $|w|$. For a tuple $\vec{w} = (w_1, \ldots, w_d)$, write $h^{(d)}(\vec{w}) := (h(w_1), \ldots, h(w_d)) \in M^d$.

Definition 2.5 (Refinement of finite typings)**.** Let $h\colon \Sigma^* \to M$ and $h'\colon \Sigma^* \to M'$ be finite homomorphisms. We say that h' *refines* h, and write $h \preceq h'$, when $\ker h' \subseteq \ker h$. Equivalently, the value of h factors through the image of h': there is a unique monoid homomorphism $\upsilon\colon \operatorname{im} h' \to \operatorname{im} h$ satisfying $h = \upsilon \circ h'$, after restricting the codomains to their images.

For two typings h and g, their product typing (h, g) refines both and satisfies $\ker(h, g) = \ker h \cap \ker g$.

Finite-monoid homomorphisms serve here as finite-state typings. Let $h_{\mathrm{triv}} : \Sigma^* \to \{1\}$ be the trivial typing. A DFA transition morphism for a regular superset of the target is another example [27]; only the induced finite summary is used, not a membership oracle. We call $h^{(d)}(\vec{w})$ the *yield type*. The prefix–suffix morphism below supplies the boundary typing used later.

Definition 2.6 (Prefix–suffix typing)**.** For $k, \ell \geq 0$, put $m_0 := \max\{1, k + \ell\}$. For a word w, let $\mathrm{pref}_k(w)$ and $\mathrm{suf}_\ell(w)$ denote its prefix and suffix of the indicated lengths whenever $|w| \geq m_0$, with the length-zero windows equal to ε. Define

$$h_\Sigma^{k,\ell}(w) = \begin{cases} w, & |w| < m_0, \\ \langle \mathrm{pref}_k(w), \mathrm{suf}_\ell(w) \rangle, & |w| \geq m_0, \end{cases}$$

using disjoint tags for the short-word and prefix–suffix elements. Put $M_{k,\ell} := h_\Sigma^{k,\ell}(\Sigma^*)$. For short words x, y and long elements $\langle u, v\rangle$, define

$$x \cdot y := h_\Sigma^{k,\ell}(xy), \qquad x \cdot \langle u, v\rangle := \langle \mathrm{pref}_k(xu), v\rangle,$$
$$\langle u, v\rangle \cdot y := \langle u, \mathrm{suf}_\ell(vy)\rangle, \quad \langle u, v\rangle \cdot \langle u', v'\rangle := \langle u, v'\rangle.$$

With this multiplication, $h_\Sigma^{k,\ell}(xy) = h_\Sigma^{k,\ell}(x) \cdot h_\Sigma^{k,\ell}(y)$. Since $h_\Sigma^{k,\ell}$ is surjective, associativity is inherited from concatenation and $h_\Sigma^{k,\ell}(\varepsilon)$ is the identity. Thus $h_\Sigma^{k,\ell} : \Sigma^* \to M_{k,\ell}$ is a finite-monoid homomorphism.

A finite typing h is called (k, ℓ)-*boundary-determined* when $h \preceq h_\Sigma^{k,\ell}$. Equivalently, equality of prefix–suffix types implies equality of h-types; a factor map exists on the image of the prefix–suffix typing.

For brevity, we call such an h a *boundary typing*.

2.3 Multiple Context-Free Grammars

We use the standard tuple-generating definition of multiple context-free grammars [30]; see also [31] for a general overview of MCFGs and related mildly context-sensitive formalisms. The general-rank notation below is used throughout the reconstruction and proofs.

Definition 2.7 (Standard MCFG presentation)**.** A *multiple context-free grammar* is a tuple $G = (V, \Sigma, P, S, \mathrm{fan})$, where V is a finite set of nonterminals, Σ is a finite terminal alphabet, $S \in V$ has $\mathrm{fan}(S) = 1$, and $\mathrm{fan}(A) \geq 1$ is the fan-out of each $A \in V$. A rule R of rank $r \geq 0$ has the form $R: \quad A \to \rho(B_1, \ldots, B_r), \qquad \rho = (\alpha_1, \ldots, \alpha_{d_0})$, where ρ is the output template, $d_0 = \mathrm{fan}(A)$, $d_j = \mathrm{fan}(B_j)$, and, with

$$\Xi_j := \{\xi_j^1, \ldots, \xi_j^{d_j}\}, \qquad \Gamma_\rho := \Sigma \uplus \Xi_1 \uplus \cdots \uplus \Xi_r,$$

each $\alpha_i \in \Gamma_\rho^*$ and every variable in $\Xi_1 \uplus \cdots \uplus \Xi_r$ occurs at most once in the complete tuple $(\alpha_1, \ldots, \alpha_{d_0})$. We call this at-most-once condition *linearity*. The rule is *nondeleting* if every such variable occurs exactly once. Thus nondeleting does not mean that derived tuple components are nonempty; empty strings remain allowed.

For child tuples $\vec{u}_j = (u_{j,1}, \ldots, u_{j,d_j}) \in (\Sigma^*)^{d_j}$, simultaneous substitution of $u_{j,\kappa}$ for ξ_j^κ defines $\rho(\vec{u}_1, \ldots, \vec{u}_r) \in (\Sigma^*)^{d_0}$. When $r = 0$, the rule is a constant rule and derives the tuple $(\alpha_1, \ldots, \alpha_{d_0})$. The tuple languages $L_A(G)$ form the least family, ordered componentwise by inclusion, that is closed under all rule applications, and $L(G) := \{w \in \Sigma^* : (w) \in L_S(G)\}$. Equivalently, a finite derivation tree rooted at A is evaluated bottom-up by the rule templates. We write $A \Rightarrow_G^* \vec{u}$ when such a tree yields $\vec{u} \in (\Sigma^*)^{\mathrm{fan}(A)}$; this is a *terminal derivation*, and $\vec{u}$ is its *terminal tuple*. A terminal derivation $S \Rightarrow_G^* (w)$ from the start symbol is called *successful*. An f-MCFG has $\mathrm{fan}(A) \leq f$ for every nonterminal, and an f-MCFL is a language generated by an f-MCFG. A grammar has *rule-rank cap* $r_{\max}$ if every rule has rank at most $r_{\max}$.

At fan-out one, the formalism includes ordinary context-free grammars. A *linear context-free grammar* is a CFG in which every production has at most one nonterminal on its right-hand side; a *linear context-free language* is generated by such a grammar. This is distinct from the "linearity" condition on MCFG rule templates above, which means that each rule variable occurs at most once.

Fix a standard explicit symbol-level encoding of finite MCFG presentations and write $\|G\|$ for its length. The encoding counts the nonterminal and rule records and every terminal or variable occurrence in each rule template. All polynomial grammar-size bounds below refer to this encoding.

For integers $f \geq 1$ and $r \geq 0$, write

$$\mathsf{MCFL}(f,r) := \{L(G) : G \text{ is an MCFG of fan-out at most } f, \text{ with rule rank at most } r\}.$$

Thus f-MCFL $= \bigcup_{r<\infty} \mathsf{MCFL}(f,r)$, because every finite MCFG presentation has finite rule rank. When minimum rule rank at fan-out f is mentioned below, it means the language-level minimum over all fan-out-f presentations, not the rank of a particular supplied grammar.

We use f throughout for fan-out/dimension and r for a generic rule rank; $r_{\max}$ denotes a prescribed rank cap. Rule-template variables are always written ξ_j^κ, so that x, y, z (with or without arrows) remain free for words and observed tuples. Terminals in examples, including digits and #, are typeset in `typewriter` font.

When two nonterminals have the same fan-out d, we use $A \to B$ as shorthand for the unary identity-template rule $A \to (\xi_1^1, \ldots, \xi_1^d)(B)$. This denotes only the identity correspondence between components: at fan-out two, for example, it means $A \to (\xi_1^1, \xi_1^2)(B)$, not a swapping rule such as $A \to (\xi_1^2, \xi_1^1)(B)$. Likewise, $A \to \vec{c}$ denotes a rank-zero constant rule. These abbreviations are used for typed start links and for Rule (Link) of the reconstruction below.

The standard formalism allows rank zero, nonidentity unary rules, arbitrary finite rule rank, and deleting rules. A nondeleting rule $R : A \to \rho(B_1, \ldots, B_r)$, with $\rho = (\alpha_1, \ldots, \alpha_{d_0})$, is *nonpermuting* if, for each child B_j, the variables $\xi_j^1, \ldots, \xi_j^{\mathrm{fan}(B_j)}$ occur in this order when the parent components are scanned from left to right. Following Yoshinaka [14, Section 2.3], a rule is *λ-free* if no output template α_i is empty, and a nonpermuting rule is *nonmerging* if no α_i contains adjacent variables $\xi_j^\kappa \xi_j^{\kappa+1}$ of the same child. A presentation is *good* if all its rules are λ-free, nondeleting, nonpermuting, and nonmerging. We write ε for the empty word throughout; "λ-free" is retained only as Yoshinaka's standard name for this rule condition. We use good presentations in the completeness proof. The nonmerging condition ensures that, after the siblings of a child are fixed, consecutive components of that child remain separated by nonempty terminal material or by a nonempty sibling yield. This matches Yoshinaka's multidimensional-context convention. In particular, λ-freeness implies by induction that every component of every terminal tuple derived from a nonterminal is nonempty.

The normalization used below uses only the nondeleting step (f3) in the proof of Seki et al.'s Lemma 2.2, in the rank-preserving form recorded by Kanazawa, together with component permutation, explicit empty-component removal, and Yoshinaka's nonmerging conversion [30, proof of Lemma 2.2, item (f3)] [32, Section 2.1] [14, Section 2.3, Lemma 1]. Yoshinaka also notes later in Section 2.3, after Example 2, that for fixed dimension and rule rank these standard conversions yield an equivalent good grammar of size $O(\|G\|)$ [14, Section 2.3].

Fan-out and rule rank are distinct grammar resources. In particular, there is no blanket binarization theorem that preserves an arbitrary fixed fan-out for general MCFG/LCFRS presentations: reducing rule rank may require increased fan-out [24, 33]. Accordingly the qualitative reconstruction below is defined without imposing a binary normal form or a known rank cap.

Reduced grammars. The nonterminal dependency graph of an MCFG has an edge $A \to B$ whenever B occurs as a child in a rule with left-hand side A. A nonterminal is *reachable* if it is reachable from the start symbol in this graph, and *productive* if its tuple language is nonempty. The grammar is *reduced* if every nonterminal is both reachable and productive.

2.4 Normalization

Proposition 2.8 (Good normalization with rank preservation)**.** *Let G_{in} be a finite standard f-MCFG of maximum rule rank at most $r_{\max}$. There is a finite good f-MCFG G_{good} of maximum rule rank*

at most $r_{\max}$ *such that* $L(G_{\text{good}}) = L(G_{\text{in}}) \setminus \{\varepsilon\}$. *For fixed* f *and* $r_{\max}$, $\|G_{\text{good}}\| = O_{f,r_{\max}}(\|G_{\text{in}}\|)$. *If* $\varepsilon \notin L(G_{\text{in}})$, *the two grammars are equivalent.*

Proof. First delete unreachable or unproductive nonterminals, then apply the nondeleting step (f3) from the proof of Seki et al.'s Lemma 2.2, in the rank-preserving form recorded by Kanazawa [30, proof of Lemma 2.2, item (f3)] [32, Section 2.1]. Decorating nonterminals by component permutations makes the grammar nonpermuting. Next decorate by nonempty-component patterns realizable by terminal derivations, as made explicit in Appendix A, and project away empty components; this removes ε at the start and makes every retained output template λ-free. Finally apply Yoshinaka's nonmerging conversion [14, Section 2.3, Lemma 1]. None of these steps increases fan-out or rule rank. With $f, r_{\max}$ fixed, component-pattern choices contribute at most 2^f nonterminal decorations and $2^{fr_{\max}}$ child-pattern choices per rule, while permutation decoration contributes at most $f!$ copies. Yoshinaka's conversion has the same fixed-parameter linear-size behavior. Hence the total encoding size remains $O_{f,r_{\max}}(\|G_{\text{in}}\|)$. The derivation correspondence and the empty-component construction are recorded in Appendix A. □

This proposition controls size, fan-out, and rule rank, but it does not identify the rule thickness of an arbitrary presentation with the thickness after normalization. Thickness-sensitive statements are therefore first proved for good presentations. For reduced λ-free input presentations, including presentations with deleting rules, Proposition A.1 gives a polynomial grammar-size-and-thickness transfer to the good presentation used below.

2.5 Tuple Contexts and Substitutability

Sentence contexts. A *sentence context of arity* d is a word $E = v_0\Box_1 v_1 \cdots v_{d-1}\Box_d v_d$ in which each named hole occurs exactly once, $v_0, v_d \in \Sigma^*$, and every internal separator $v_1, \ldots, v_{d-1}$ belongs to Σ^+. Let $\mathcal{E}_d$ be the set of such contexts. This is Yoshinaka's multidimensional-context convention [14, Section 3.1]. For $\vec{w} = (w_1, \ldots, w_d)$, write $E[\vec{w}]$ for the result of filling $\Box_i$ by w_i, and put $|E|_{\text{fix}} := \sum_{i=0}^{d} |v_i|$. For brevity, we say that a sentence context E *accepts* $\vec{w}$ in a language L when $E[\vec{w}] \in L$.

Definition 2.9 (Tuple occurrence in a sample word)**.** Let $K \subseteq \Sigma^*$ and $w = a_1 \cdots a_n \in K$. Number the cut positions of w as $⓪a_1① \cdots a_n⓷$. For $0 \le i < i' \le n$, write $w[i : i'] := a_{i+1} \cdots a_{i'}$.

An *arity-d tuple occurrence* in w is a sequence of nonempty, left-to-right intervals $0 \le i_1 < i'_1 < i_2 < i'_2 < \cdots < i_d < i'_d \le n$. It determines the tuple $\vec{x} = (w[i_1 : i'_1], \ldots, w[i_d : i'_d]) \in (\Sigma^+)^d$ and the context $E \in \mathcal{E}_d$ obtained by replacing those intervals by $\Box_1, \ldots, \Box_d$. We identify the occurrence with $(E, \vec{x})$. For example, the two intervals spelling `a` and `c` in `abc` give the tuple $(\mathtt{a}, \mathtt{c})$ and the context $\Box_1\mathtt{b}\Box_2$.

For $L \subseteq \Sigma^*$ and $\vec{x} \in (\Sigma^*)^d$, define its *d-tuple distribution* $\mathcal{D}_L^{(d)}(\vec{x}) := \{E \in \mathcal{E}_d : E[\vec{x}] \in L\}$. When d is clear, the superscript is omitted.

Definition 2.10 ((f,h)-tuple substitutability)**.** Let $f \ge 1$ and let $h : \Sigma^* \to M$ be a finite monoid homomorphism. A language $L \subseteq \Sigma^*$ is (f,h)*-tuple-substitutable* if, for every $1 \le d \le f$ and all $\vec{x}, \vec{y} \in (\Sigma^+)^d$,

$$h^{(d)}(\vec{x}) = h^{(d)}(\vec{y}), \qquad \mathcal{D}_L^{(d)}(\vec{x}) \cap \mathcal{D}_L^{(d)}(\vec{y}) \neq \emptyset$$

imply $\mathcal{D}_L^{(d)}(\vec{x}) = \mathcal{D}_L^{(d)}(\vec{y})$. Thus empty strings are not compared as tuple components; when ε belongs to the target, it is handled separately at the start symbol. This is Yoshinaka's nonempty-component convention for multidimensional substitutability [14, Section 3.1 and footnote 3]; see also Clark [8]

for the CFG case. The restriction is substantive already at fan-out one: if ε itself were allowed among the compared fragments, the ordinarily substitutable language a^+ would fail the condition because a and ε share a context but do not have the same distribution.

At fan-out one, every sentence context has the form $u\Box_1 v$, so $\mathcal{D}_L^{(1)}(x)$ is canonically the ordinary two-sided distribution $\mathcal{D}_L(x) = \{(u, v) : uxv \in L\}$. Hence Definition 2.10 specializes, without any external convention, to the ordinary two-sided fixed-h substitutability condition on nonempty factors.

Typing–distribution equivalence. For $\vec{x}, \vec{y} \in (\Sigma^+)^d$, write $\vec{x} \equiv_{L,h}^d \vec{y}$ if and only if

$$h^{(d)}(\vec{x}) = h^{(d)}(\vec{y}) \qquad \text{and} \qquad \mathcal{D}_L^{(d)}(\vec{x}) = \mathcal{D}_L^{(d)}(\vec{y}).$$

For fixed L, h, d this is an equivalence relation; the superscript is omitted when the arity is clear.

Definition 2.11 (The target class)**.** Fix $f \geq 1$ and a finite monoid homomorphism h. The class $\mathcal{C}_{f,h}^{\mathrm{mcf}}$ consists of all f-MCFLs that are (f, h)-tuple-substitutable.

A target $L \in \mathcal{C}_{f,h}^{\mathrm{mcf}}$ is called *boundary-typed* when h is a boundary typing.

If $h \preceq h'$, then $\mathcal{C}_{f,h}^{\mathrm{mcf}} \subseteq \mathcal{C}_{f,h'}^{\mathrm{mcf}}$: equality of the finer h'-types implies equality of the coarser h-types, so the substitution conclusion available under h applies. At fan-out one this is the context-free part of the two-sided fixed-h condition just described.

For every $f \geq 1$, $\mathcal{C}_{f+1,h}^{\mathrm{mcf}} \cap f\text{-MCFL} \subseteq \mathcal{C}_{f,h}^{\mathrm{mcf}}$, because the fan-out-$f$ condition tests only the arities $1, \ldots, f$. The possible loss of substitutability when the tested arity is increased is already present for trivial typing: Yoshinaka notes that $\mathsf{SL}(p, r)$ and $\mathsf{SL}(p + 1, r)$ are generally incomparable and gives $L_{\mathrm{reverse}} = \{w\#w^R\}$ as a one-dimensional substitutable language that is not two-dimensional substitutable [14, Section 4.1 and Example 5]. What finite typing adds here is the stronger phenomenon of Proposition 5.3: a target can be learnable at fan-out one under a finite typing yet fail the fan-out-two condition for *every* finite typing. Thus f is part of the learning bias, not merely an upper bound on representational power. No monotonicity of the learned classes in f is claimed.

For every $\mathsf{Acc} \subseteq M$, the language $h^{-1}(\mathsf{Acc})$ belongs to $\mathcal{C}_{f,h}^{\mathrm{mcf}}$. Indeed, it is regular, and equal componentwise h-types give the same total h-value in every sentence context, hence equal tuple distributions.

Proposition 2.12 (Intersection and abelian filtering)**.** *Fix f and h.*

(i) If L_1 and L_2 are (f, h)-tuple-substitutable, then $L_1 \cap L_2$ is (f, h)-tuple-substitutable.

(ii) Let L be (f, h)-tuple-substitutable, let $\mathbb{A}$ be an abelian group, let $\varphi : \Sigma^* \to \mathbb{A}$ be a monoid homomorphism, let $\mathcal{H} \leq \mathbb{A}$, and let $\gamma \in \mathbb{A}$. Then $L \cap \varphi^{-1}(\gamma + \mathcal{H})$ is (f, h)-tuple-substitutable.

Whenever the resulting intersection is an f-MCFL, it therefore belongs to $\mathcal{C}_{f,h}^{\mathrm{mcf}}$.

Proof. For (i), equal-type tuples $\vec{x}, \vec{y}$ with some context E satisfying $E[\vec{x}], E[\vec{y}] \in L_1 \cap L_2$ have equal tuple distributions in each L_i, hence in their intersection. For (ii), first observe that $\varphi^{-1}(\gamma + \mathcal{H})$ is (f, h_{triv})-tuple-substitutable. Indeed, if a common context accepts tuples $\vec{x}, \vec{y}$, then $\sum_i(\varphi(y_i) - \varphi(x_i)) \in \mathcal{H}$. By commutativity, for every other sentence context E' the difference $\varphi(E'[\vec{y}]) - \varphi(E'[\vec{x}])$ is the same element of $\mathcal{H}$, so coset membership is preserved in both directions. Trivial typing is coarser than h, hence the same coset language is (f, h)-tuple-substitutable. Apply (i). □

As an immediate consequence, for any $\mathsf{Acc} \subseteq M$, abelian group $\mathbb{A}$, homomorphism $\varphi : \Sigma^* \to \mathbb{A}$, subgroup $\mathcal{H} \leq \mathbb{A}$, and $\gamma \in \mathbb{A}$, every f-MCFL of the form $h^{-1}(\mathsf{Acc}) \cap \varphi^{-1}(\gamma + \mathcal{H})$ belongs to $\mathcal{C}_{f,h}^{\mathrm{mcf}}$. Thus one typing can support a family of targets while the abelian constraint varies; these factorizations are not supplied to the learner.

3 Exact Reconstruction from Positive Data

We define the finite-sample reconstruction operator, prove soundness, and then prove completeness using a normalized target grammar refined by yield type. The target grammar is used only to select a finite characteristic sample. Since every observed tuple component is nonempty, the rank of an enumerated composition witness is bounded by the length of the sample word containing it.

3.1 Set-Driven Reconstruction and Sample-Bounded Rank

Following Definition 2.9, enumerate all tuple occurrences of arity at most f in the sample K. The reconstruction uses one nonterminal $[\vec{x}]$ for every observed tuple value $\vec{x} \in (\Sigma^+)^d$, together with a fresh start symbol $\widehat{S}$. It is not told which occurrences correspond to target nonterminals or rules; all admissible segmentations are enumerated.

Definition 3.1 (Composition witness)**.** Let $r \geq 1$. A *composition witness of rank r* in K consists of an observed parent occurrence $(E_{\mathrm{par}}, \vec{z})$ of arity $e \leq f$ together with a segmentation of the parent components into terminal gaps and nonempty labelled intervals. The labels are variables ξ_j^κ with $1 \leq j \leq r$ and $1 \leq \kappa \leq d_j \leq f$; every variable occurs exactly once, and for each fixed j the variables $\xi_j^1, \ldots, \xi_j^{d_j}$ occur in this order when the parent components are scanned from left to right. No output component may contain adjacent variables $\xi_j^\kappa \xi_j^{\kappa+1}$ of the same child. The labelled intervals induce child tuples $\vec{u}_j = (u_{j,1}, \ldots, u_{j,d_j})$ and determine a unique linear template ρ satisfying $\vec{z} = \rho(\vec{u}_1, \ldots, \vec{u}_r)$.

Lemma 3.2 (Materialization of child tuples)**.** *Every child tuple induced by a composition witness is itself an observed tuple occurrence, and the induced template ρ is good.*

Proof. Because the parent is an observed tuple occurrence, the components of each $\vec{u}_j$ occur in their original left-to-right order in the same sample word. If two consecutive components of the same child lie in one parent component, the definition of a composition witness excludes their direct adjacency. They are therefore separated either by nonempty terminal material or by one or more nonempty labelled intervals belonging to other children. If they lie in different parent components, they are separated by the nonempty internal material of the parent sentence context. Hence the components of each $\vec{u}_j$ satisfy the separation condition for an observed tuple occurrence.

Every variable occurs exactly once, the components of each child occur in order, and adjacent consecutive components of one child are excluded. Therefore ρ is linear, nondeleting, nonpermuting, and nonmerging. Finally every parent component is nonempty, so every output component of ρ is nonempty. Thus ρ is good. □

Lemma 3.3 (Enumeration and sample-bounded rank)**.** *Fix f and a finite sample K, and put $n_K := \max\{1, \|K\|_+\}$. If a rank-$r$ composition witness has parent tuple $\vec{z}$ occurring in a sample word w, then $r \leq \|\vec{z}\|_1 \leq |w| \leq \|K\|_+$. All composition witnesses are effectively enumerable. Under a rule-rank cap $r_{\max}$, those of rank at most $r_{\max}$ are enumerable in $n_K^{O(fr_{\max})}$ time and space.*

Proof. Every child tuple has at least one nonempty component. Linearity and nondeletion make the child components contribute disjointly to the parent, so $\|\vec{z}\|_1 \geq \sum_j \|\vec{u}_j\|_1 \geq r$; the remaining inequalities follow because the parent components are disjoint intervals of w.

An arity-d tuple occurrence is determined by $2d$ cuts. For a fixed rank r, a candidate segmentation has at most fr labelled child intervals, whose endpoints are chosen among the $|w| + 1 \leq n_K + 1$ cut positions of the parent sample word. For $r \leq r_{\max}$, assignments to parent components and terminal gaps contribute only a factor depending on f and $r_{\max}$, and the nonpermuting and nonmerging

conditions are directly checkable. Summing over $r \le r_{\max}$ gives the stated bound. Without a cap, the displayed rank inequality restricts the effective enumeration to $1 \le r \le \|K\|_+$. □

Definition 3.4 (Capped and uncapped reconstruction operators)**.** Fix f and h. For $r_{\max} \ge 1$, let $\mathcal{B}_{f,h}^{\le r_{\max}}(K)$ be the grammar $\widehat{G}_{\le r_{\max}}(K)$ with a fresh start symbol $\widehat{S}$, one nonterminal $[\vec{x}]$ for each nonempty tuple value $\vec{x}$ observed in K, and the following rules:

(Start) for every $w \in K \cap \Sigma^+$, $\widehat{S} \to [(w)]$, and if $\varepsilon \in K$, also $\widehat{S} \to \varepsilon$;

(Const) for every observed tuple, the constant rule $[\vec{u}] \to \vec{u}$;

(Comp) for every composition witness of rank $1 \le r \le r_{\max}$ with parent tuple $\vec{z}$, template ρ, and induced child tuples $\vec{u}_1, \dots, \vec{u}_r$, the rule $[\vec{z}] \to \rho([\vec{u}_1], \dots, [\vec{u}_r])$;

(Link) for observed nonempty tuples $\vec{x}, \vec{y}$ of the same arity d, the identity-template rule $[\vec{x}] \to [\vec{y}]$ whenever $h^{(d)}(\vec{x}) = h^{(d)}(\vec{y})$ and some $E \in \mathcal{E}_d$ satisfies $E[\vec{x}], E[\vec{y}] \in K$.

If $K \ne \emptyset$, define $\mathcal{B}_{f,h}(K) := \widehat{G}(K) := \widehat{G}_{\le \|K\|_+}(K)$; for $K = \emptyset$, let $\mathcal{B}_{f,h}(\emptyset)$ be the grammar with no productive start rule. Fixed encoding orders make both reconstruction operators deterministic and set-driven.

Exact relation to Yoshinaka's finite-sample grammar. The finite typing modifies Yoshinaka's constructor only at substitution links.

Proposition 3.5 (Typed guard on Yoshinaka's reconstruction)**.** *Fix a finite positive sample K, fan-out bound f, and rank cap $r_{\max} \ge 1$. Under the trivial typing, $\mathcal{B}_{f,h_{\mathrm{triv}}}^{\le r_{\max}}(K)$ agrees, up to notation, with Yoshinaka's hypothesis grammar $G(K)$ for $\mathcal{A}(f, r_{\max})$ [14, Section 4.1], except that the present construction handles $\varepsilon \in K$ by a separate direct start rule. For an arbitrary finite typing h, $\mathcal{B}_{f,h}^{\le r_{\max}}(K)$ is obtained from that grammar by deleting exactly those Type II substitution rules whose source and target tuples have different componentwise h-types. The same statement holds for the uncapped reconstruction and Yoshinaka's $\mathcal{A}(f, *)$.*

Proof. The observed tuple values are the same nonterminal indices in both constructions. Rule (Start) is Yoshinaka's Type III rule, and Rule (Link) is Type II with the additional componentwise type test.

It remains to compare Type I with Rules (Const) and (Comp). A rank-zero Type I rule is exactly Rule (Const). For positive rank, suppose Yoshinaka has

$$[\vec{z}] \to \rho([\vec{u}_1], \dots, [\vec{u}_r]) \qquad \text{with} \qquad \vec{z} = \rho(\vec{u}_1, \dots, \vec{u}_r),$$

where all displayed tuple values are observed and ρ is good. Choose an observed occurrence of $\vec{z}$ in a sample word. The displayed equality locates every component of every $\vec{u}_j$ as a labelled interval inside that occurrence. Linearity and nondeletion use each interval once, nonpermutation puts the components of each child in order, and nonmerging prevents direct adjacency of consecutive components of one child. Hence these intervals form a composition witness, so Rule (Comp) contains the same rule. Conversely every composition witness is a good algebraic decomposition among observed tuple values and therefore is a Type I rule. This also matches Yoshinaka's counting proof, where each Type I rule is enumerated as a factorization of a single sample word [14, Lemma 6].

Thus trivial typing gives the same Type I–III rule set, apart from the direct ε start convention. For general h, Rules (Start), (Const), and (Comp) are unchanged, and only Type II/Rule (Link) is filtered by componentwise type. Finally, in the uncapped construction every composition witness has rank at most the length of the sample word containing it, so allowing the cap $\|K\|_+$ introduces no Type I rule beyond the sample-length cap used in Yoshinaka's $\mathcal{A}(f, *)$. □

Example 3.6 (Why the equal-type premise matters). Let $L = \{\mathtt{a}^n\mathtt{b}^n\mathtt{c}^n : n \geq 1\}$ and $K = \{\mathtt{abc}, \mathtt{aabbcc}\} \subseteq L$. The observed factors $(\mathtt{b})$ and $(\mathtt{abbc})$ share the sample context $\mathtt{a}\square_1\mathtt{c}$. With trivial typing, Rule (Link) therefore connects their hypothesis nonterminals. The composition witness $[\mathtt{aabbcc}] \to \mathtt{aa}[\mathtt{b}]\mathtt{bcc}$ can then use that link to derive $\mathtt{aaabbcbcc} \notin L$. Under $h_\Sigma^{1,1}$ the link is blocked because $h_\Sigma^{1,1}(\mathtt{b}) = \mathtt{b}$ while $h_\Sigma^{1,1}(\mathtt{abbc}) = \langle \mathtt{a}, \mathtt{c}\rangle$. This illustrates the local role of the equal-type premise.

The reconstruction is sample-consistent:

$$K \subseteq L(\mathcal{B}_{f,h}^{\leq r_{\max}}(K)), \qquad K \subseteq L(\mathcal{B}_{f,h}(K)).$$

Indeed, each nonempty $w \in K$ has the direct derivation $\widehat{S} \to [(w)] \to (w)$ by Rules (Start)–(Const), while $\varepsilon \in K$ gives the direct start rule $\widehat{S} \to \varepsilon$.

These $\mathcal{B}$ operators are set-driven constructors, not sequential TxtEx learners: recomputation on growing samples need not stabilize syntactically (e.g. for successively longer $\mathtt{a}^n\mathtt{cb}^n$). The conservative learner below rebuilds only when a new positive datum is rejected.

Both reconstruction operators produce finite standard MCFGs of fan-out at most f and are effectively constructible from K, f, h. There are only finitely many observed nonempty tuple occurrences. Rule (Link) is obtained by grouping occurrences by concrete context and componentwise type, while Lemma 3.3 enumerates Rule (Comp). By Lemma 3.3, every rule of $\mathcal{B}_{f,h}(K)$ has rank at most $\|K\|_+$ when $K \neq \emptyset$.

The concrete separation examples in Section 5 also serve as small instances of the tuple-nonterminal and composition mechanisms used here.

3.2 Soundness

Lemma 3.7 (Induced child context)**.** *Let ρ be the template of a good rule, let $E \in \mathcal{E}_e$ be a sentence context for its output tuple, and fix tuples of the arities required by the child nonterminals, with nonempty components, for all children except child j. Replacing every variable of child j by its named hole and every other child variable by the corresponding fixed component produces a context $E_j \in \mathcal{E}_{d_j}$ such that $E_j[\vec{u}_j] = E[\rho(\vec{u}_1, \ldots, \vec{u}_r)]$. If the sibling tuples are changed, the same construction gives the corresponding induced context E_j for that new sibling choice; the context itself need not be identical.*

Proof. Linearity and nondeletion make every variable of child j occur exactly once, and nonpermutingness puts $\xi_j^1, \ldots, \xi_j^{d_j}$ in that order. Between two consecutive variables of this child, nonmerging ensures either a terminal symbol, a variable of another child, or a boundary between two output components. In the first case the separator is visibly nonempty; in the second it becomes a nonempty sibling component; in the third the corresponding internal separator of E is nonempty. Replacing the distinguished variables by $\square_1, \ldots, \square_{d_j}$ therefore gives a context in $\mathcal{E}_{d_j}$, and filling the holes is exactly composition with ρ and then with E. □

Lemma 3.8 (Composition preserves equivalence)**.** *Let $L \subseteq \Sigma^*$, let ρ be the template of a good rank-r rule, and let $\vec{x}_j, \vec{u}_j$ have the arities required by its children and have nonempty components. If $\vec{u}_j \equiv_{L,h} \vec{x}_j$ for every j, then $\rho(\vec{u}_1, \ldots, \vec{u}_r) \equiv_{L,h} \rho(\vec{x}_1, \ldots, \vec{x}_r)$.*

Proof. Homomorphic template evaluation gives equality of the two parent h-types, and goodness of ρ makes every parent component nonempty. It remains to compare distributions. Let E be an arbitrary sentence context for the parent tuple. If $E[\rho(\vec{x}_1, \ldots, \vec{x}_r)] \in L$, replace the children one at a time. At step j, Lemma 3.7 supplies the induced sentence context for child j under the

current sibling choice. Equality of the child distributions transfers acceptance from $\vec{x}_j$ to $\vec{u}_j$. After all replacements, $E[\rho(\vec{u}_1, \ldots, \vec{u}_r)] \in L$. Reversing the replacements gives the converse implication. Since E was arbitrary, the two parent tuples have equal distributions. □

Proposition 3.9 (Soundness of every capped hypothesis)**.** *Let L be (f,h)-tuple-substitutable and $K \subseteq L$. For every $r_{\max} \geq 1$, if $[\vec{x}] \Rightarrow^* \vec{u}$ in $\widehat{G}_{\leq r_{\max}}(K)$, then every component of $\vec{u}$ is nonempty and $\vec{u} \equiv_{L,h} \vec{x}$. Consequently*

$$L(\mathcal{B}_{f,h}^{\leq r_{\max}}(K)) \subseteq L, \qquad L(\mathcal{B}_{f,h}(K)) \subseteq L.$$

Proof. Induct on derivation height. Constant rules are immediate. For a Rule (Link) instance $[\vec{x}] \to [\vec{y}]$, its defining common sample context and equal h-type give $\vec{x} \equiv_{L,h} \vec{y}$; transitivity with the induction hypothesis proves the claim. For a composition-witness rule, the induction hypotheses give nonempty child tuples equivalent to the witnessed children. Every parent component of the witnessed tuple was nonempty; under the same nondeleting template it remains nonempty after replacing child components by other nonempty strings. Lemma 3.8 then gives the parent equivalence.

A nonempty start derivation begins with $\widehat{S} \to [(w_0)]$ for some $w_0 \in K \cap \Sigma^+$, so the empty unary context places the derived word in L. The direct rule $\widehat{S} \to \varepsilon$ is present only when $\varepsilon \in K \subseteq L$. The $K = \emptyset$ case is trivial, and the general hypothesis is the capped case with $r_{\max} = \|K\|_+$ whenever $K \neq \emptyset$. □

3.3 Yield-Typed Target Refinement and Finite Witnesses

Definition 3.10 (Template yield type)**.** Let $R : A \to \rho(B_1, \ldots, B_r)$ be a nondeleting rule with output template $\rho = (\alpha_1, \ldots, \alpha_{d_0})$, and put $d_0 = \mathrm{fan}(A)$ and $d_j := \mathrm{fan}(B_j)$, and choose $\mathbf{t}_j \in M^{d_j}$ for each $1 \leq j \leq r$.

On the template alphabet $\Gamma_\rho := \Sigma \uplus \biguplus_{j=1}^{r} \{\xi_j^1, \ldots, \xi_j^{d_j}\}$, define $\mathsf{ev}_{\mathbf{t}_1,\ldots,\mathbf{t}_r} : \Gamma_\rho \to M$ by $\mathsf{ev}_{\mathbf{t}_1,\ldots,\mathbf{t}_r}(a) := h(a)$ for $a \in \Sigma$, and $\mathsf{ev}_{\mathbf{t}_1,\ldots,\mathbf{t}_r}(\xi_j^\kappa) := (\mathbf{t}_j)_\kappa$.

Let $\widehat{\mathsf{ev}}_{\mathbf{t}_1,\ldots,\mathbf{t}_r} : \Gamma_\rho^* \to M$ be its unique extension to a monoid homomorphism. Define

$$\mathrm{out}_\rho(\mathbf{t}_1, \ldots, \mathbf{t}_r) := (\widehat{\mathsf{ev}}_{\mathbf{t}_1,\ldots,\mathbf{t}_r}(\alpha_1), \ldots, \widehat{\mathsf{ev}}_{\mathbf{t}_1,\ldots,\mathbf{t}_r}(\alpha_{d_0})) \in M^{d_0}.$$

For rank zero, out_ρ is simply $(h(\alpha_1), \ldots, h(\alpha_{d_0}))$, the componentwise h-type of the constant template.

The refinement below is ordinary finite nonterminal splitting: compositionality $h(uv) = h(u)h(v)$ makes each typed target nonterminal uniform in componentwise h-type, which is exactly the guard needed for substitutability.

Definition 3.11 (Yield-type refinement)**.** Given a finite nondeleting MCFG $G = (V, \Sigma, P, S, \mathrm{fan})$ and a finite typing h, the *full yield-type refinement* G^h has a fresh start symbol $\widetilde{S}$ and typed nonterminals $A_{\mathbf{t}}$ for every $A \in V$ and $\mathbf{t} \in M^{\mathrm{fan}(A)}$. Set $\mathrm{fan}(\widetilde{S}) := 1$ and $\mathrm{fan}(A_{\mathbf{t}}) := \mathrm{fan}(A)$.

For each $\eta \in M$, add the start rule $\widetilde{S} \to S_{(\eta)}$. For each rank-zero rule $A \to \vec{c}$, add $A_{h^{(\mathrm{fan}(A))}(\vec{c})} \to \vec{c}$.

For every positive-rank rule $R : A \to \rho(B_1, \ldots, B_r)$, $\rho = (\alpha_1, \ldots, \alpha_{d_0})$, and every choice $\mathbf{t}_j \in M^{\mathrm{fan}(B_j)}$ $(1 \leq j \leq r)$, add

$$A_{\mathrm{out}_\rho(\mathbf{t}_1,\ldots,\mathbf{t}_r)} \to \rho(B_{1,\mathbf{t}_1}, \ldots, B_{r,\mathbf{t}_r}).$$

Here $B_{j,\mathbf{t}_j}$ denotes the typed copy of B_j indexed by $\mathbf{t}_j$.

Finally, the *trimmed refinement* $\widetilde{G}_0$ is obtained by keeping only the typed nonterminals and typed rules that actually occur in some successful derivation from $\widetilde{S}$; the subscript 0 denotes this productive/reachable core. From this point on, a "typed nonterminal" or "typed rule" means one of $\widetilde{G}_0$ unless the full refinement G^h is mentioned explicitly.

Proposition 3.12 (Yield-type invariants)**.** *For G^h and $\widetilde{G}_0$:*

(i) if $A_{\mathbf{t}} \Rightarrow^* \vec{u}$, then $h^{(\mathrm{fan}(A))}(\vec{u}) = \mathbf{t}$, and erasing the type indices gives a valid G-derivation of $\vec{u}$;

(ii) every terminal derivation $A \Rightarrow_G^* \vec{u}$ lifts uniquely to a derivation $A_{h^{(\mathrm{fan}(A))}(\vec{u})} \Rightarrow_{G^h}^* \vec{u}$. In particular, every successful derivation $S \Rightarrow_G^* w$ lifts uniquely through the start rule $\widetilde{S} \to S_{(h(w))}$;

(iii) $L(\widetilde{G}_0) = L(G^h) = L(G)$.

Proof. At a rule application, homomorphic evaluation of the rule template gives

$$h^{(\mathrm{fan}(A))}(\rho(\vec{u}_1, \ldots, \vec{u}_r)) = \mathrm{out}_\rho(h^{(\mathrm{fan}(B_1))}(\vec{u}_1), \ldots, h^{(\mathrm{fan}(B_r))}(\vec{u}_r)).$$

The rank-zero case is immediate. Induction on derivation height proves (i), and the same identity gives the unique terminal-derivation lifting in (ii). Erasing the type indices gives one language inclusion. Conversely, every successful terminal derivation of G has the unique typed lifting described in (ii), including the uniquely determined start link $\widetilde{S} \to S_{(h(w))}$. Finally, every successful typed derivation consists entirely of rules retained by the trimming, so trimming preserves the generated language. □

Yield typing changes only nonterminal labels, so every typed rule has the same rule template as its untyped source. Hence λ-freeness, nondeletion, nonpermutation, and nonmerging are all inherited by the trimmed refinement.

For a typed nonterminal $X = A_{\mathbf{t}}$, put $L_X := \{\vec{u} : X \Rightarrow_{\widetilde{G}_0}^* \vec{u}\}$. Trimming makes L_X nonempty; if the target grammar is λ-free, every component of every $\vec{u} \in L_X$ is nonempty.

Lemma 3.13 (Canonical contexts for typed nonterminals)**.** *Assume the target grammar is good. Every typed nonterminal X of arity d has a sentence context $\chi(X) \in \mathcal{E}_d$ such that $\chi(X)[\vec{u}] \in L(G)$ for every $\vec{u} \in L_X$. If $\widetilde{S} \to X$ belongs to $\widetilde{G}_0$, one may take $\chi(X) = \Box_1$.*

Proof. Choose a successful typed derivation containing an occurrence of X. At the root, the one-hole context $\Box_1$ is valid. Follow the unique path from the root occurrence down to the chosen occurrence of X. At each rule on this path, fix the productive sibling subderivations used in the successful derivation and apply Lemma 3.7. Goodness of the rule and nonemptiness of the sibling components ensure that the induced child context again has nonempty internal separators. Iterating down the path gives $\chi(X) \in \mathcal{E}_d$. The surrounding typed rule applications depend only on the typed symbol X, so replacing its subtree by any other X-derivation remains a valid derivation in $\widetilde{G}_0$. If $\widetilde{S} \to X$ is a rule of $\widetilde{G}_0$, the path has length zero and $\chi(X) = \Box_1$. □

When $L = L(G) \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$, the refinement is also distributionally homogeneous for the sentence-context family used here. If $X = A_{\mathbf{t}}$ has arity d and $\vec{u}, \vec{v} \in L_X$, Proposition 3.12 gives $h^{(d)}(\vec{u}) = h^{(d)}(\vec{v}) = \mathbf{t}$, while Lemma 3.13 supplies one sentence context accepting both. Hence (f, h)-tuple substitutability gives $\mathcal{D}_L^{(d)}(\vec{u}) = \mathcal{D}_L^{(d)}(\vec{v})$. Thus each typed nonterminal lies inside one distributional class for $\mathcal{E}_d$. This parallels, but is weaker than, the full-multicontext homogeneity used for congruential MCFGs [17]: our internal separators are required to be nonempty. For genuine tuple arity $d \geq 2$, the present sentence-context family still requires nonempty internal separators, so we do not assert an inclusion into the full multicontext congruential MCFG class; whether the empty-separator cases can be recovered uniformly remains open.

Definition 3.14 (Canonical tuples and contexts)**.** For every non-start typed nonterminal X, fix a *canonical tuple* $\omega(X) \in L_X$ and a *canonical context* $\chi(X)$ supplied by Lemma 3.13. Whenever $\widetilde{S} \to X$ belongs to $\widetilde{G}_0$, require $\chi(X) = \square_1$; otherwise choose any canonical context from the lemma. For a λ-free target grammar every component of $\omega(X)$ is nonempty, so $\|\omega(X)\|_1 \geq \mathrm{fan}(X) \geq 1$. Qualitatively a canonical tuple may be the first tuple in an effective enumeration of finite terminal derivations; the quantitative sections choose short canonical tuples. These canonical objects are target witnesses, not inputs to the reconstruction operator.

Definition 3.15 (Finite target witness set)**.** Let $\mathsf{W}(\widetilde{G}_0)$ consist of the following positive words:

$\chi(X)[\omega(X)]$	for every non-start typed nonterminal X,
$\chi(X)[\rho(\omega(Y_1), \ldots, \omega(Y_r))]$	for every non-start positive-rank typed rule $X \to \rho(Y_1, \ldots, Y_r)$,
$\chi(X)[\vec{c}]$	for every non-start rank-zero typed rule $X \to \vec{c}$.

Because $\widetilde{G}_0$ is finite and all of its symbols and rules occur in successful derivations, $\mathsf{W}(\widetilde{G}_0)$ is a finite subset of $L(G)$; when $L(G) \neq \emptyset$ it is nonempty.

Lemma 3.16 (Target rule words occur in the enumeration)**.** *Let $R : X \to \rho(Y_1, \ldots, Y_r)$ be a non-start positive-rank typed rule. Put $\vec{z}_R := \rho(\omega(Y_1), \ldots, \omega(Y_r))$. If $\mathsf{W}(\widetilde{G}_0) \subseteq K$, the witness enumeration contains $[\vec{z}_R] \to \rho([\omega(Y_1)], \ldots, [\omega(Y_r)])$.*

Proof. The word $\chi(X)[\vec{z}_R] \in K$ supplies the parent occurrence. The segmentation of this sample occurrence induced by the variables of ρ places every canonical tuple of each child as a tuple occurrence in the same sample word. λ-freeness makes all child components nonempty, nonpermutingness puts each child's components in index order, and nonmerging guarantees a nonempty separator between consecutive components of each child in the sample word. Thus every child canonical tuple is an observed tuple occurrence and the segmentation is a composition witness; exhaustive enumeration emits the displayed rule. □

Proposition 3.17 (Cardinality of the finite target witness set)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be the normalized target presentation from which $\widetilde{G}_0$ is constructed, with fan-out at most f and rule rank at most $r_{\max}$. For a fixed finite typing $h : \Sigma^* \to M$, the target witness set constructed above may be chosen so that $|\mathsf{W}(\widetilde{G}_0)| \leq |V||M|^f + |P||M|^{f r_{\max}}$. In particular, for fixed f, $r_{\max}$, and h, the number of selected positive words is linear in $|V| + |P|$.*

Proof. The yield-type refinement has at most $|V||M|^f$ non-start typed nonterminals. The source rules give at most $|P||M|^{f r_{\max}}$ non-start typed rule instances. The construction selects at most one word for each such typed nonterminal and one for each such typed rule instance; fresh start links are handled separately and require no additional witness word. □

Proposition 3.17 controls only the number of selected words, not their total positive length $\|\mathsf{W}(\widetilde{G}_0)\|_+$. Polynomial total-length bounds require the additional rank-one or boundary-determined arguments below.

3.4 Completeness and Exact Reconstruction

Proposition 3.18 (Completeness from a target presentation)**.** *This is the finite-typed counterpart of Yoshinaka's rule-witness completeness argument [14, Lemma 5]. Let G be a finite good MCFG of fan-out at most f, let $L_G := L(G)$, and construct its trimmed yield-type refinement $\widetilde{G}_0$ and witness set above. Let r_G be the maximum rank of a non-start typed rule, with $r_G := 0$ if there is none. For every finite sample K with $\mathsf{W}(\widetilde{G}_0) \subseteq K$ and every $r_{\max} \geq \max\{1, r_G\}$, $L_G \subseteq L(\mathcal{B}_{f,h}^{\leq r_{\max}}(K))$, and also $L_G \subseteq L(\mathcal{B}_{f,h}(K))$.*

Proof. For each non-start typed nonterminal X, put $\iota(X) := [\omega(X)]$. If $\widetilde{S} \to X$ belongs to $\widetilde{G}_0$, then $\chi(X) = \square_1$, so the reconstruction contains the corresponding start rule.

For a non-start rank-zero typed rule $R : X \to \vec{c}$, the two corresponding witness words share $\chi(X)$, and Proposition 3.12 gives the same componentwise h-type to $\omega(X)$ and $\vec{c}$. Hence the reconstruction contains $\iota(X) \to [\vec{c}]$, followed by the constant rule.

For a non-start positive-rank typed rule $R : X \to \rho(Y_1, \ldots, Y_r)$, put $\vec{z}_R := \rho(\omega(Y_1), \ldots, \omega(Y_r))$. Yield typing gives $h^{(\mathrm{fan}(X))}(\vec{z}_R) = h^{(\mathrm{fan}(X))}(\omega(X))$, and the two corresponding witness words share $\chi(X)$, so Rule (Link) $\iota(X) \to [\vec{z}_R]$ is present. Lemma 3.16 provides $[\vec{z}_R] \to \rho(\iota(Y_1), \ldots, \iota(Y_r))$. Induction on a $\widetilde{G}_0$-derivation therefore simulates every non-start typed rule whenever the cap is at least r_G; the fresh start link was handled above. Hence every word of L_G is generated. For the uncapped reconstruction, each required rule has a witness word in K; Lemma 3.3 places its rank below $\|K\|_+$. This simulation uses neither tuple substitutability nor $K \subseteq L$; those enter only through soundness. □

Theorem 3.19 (Exact reconstruction for a fixed typing)**.** *Fix $f \geq 1$ and a finite homomorphism $h : \Sigma^* \to M$. For every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$ there is a finite characteristic sample $K^\star_L \subseteq L$ such that every finite $K^\star_L \subseteq K \subseteq L$ satisfies $L(\mathcal{B}_{f,h}(K)) = L$.*

Proof. Put $L^+ := L \setminus \{\varepsilon\}$. If $L^+ = \emptyset$, take $K^\star_L = L$. Otherwise choose a finite f-MCFG presentation of L and apply Proposition 2.8 to obtain a good presentation of L^+. Let $\widetilde{G}_0$ be its trimmed yield-type refinement and set $K^\star_L := \mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$. The observation following Definition 3.15 makes this a finite subset of L. For $K^\star_L \subseteq K \subseteq L$, Proposition 3.18 gives every nonempty target word in the hypothesis, while Proposition 3.9 gives the reverse inclusion. If $\varepsilon \in L$, the direct start rule generates it; if $\varepsilon \notin L$, soundness forbids it. The cases $L = \emptyset$ and $L = \{\varepsilon\}$ are immediate from the reconstruction definition. □

The set-driven map that sends a finite text prefix to $\mathcal{B}_{f,h}$ of its positive content is eventually semantically correct: once the content contains $K^\star_L$, every later reconstruction generates L by Theorem 3.19. Its grammar presentation may still change as new positive examples enlarge the observed tuple inventory, which is why TxtEx stabilization is handled by the conservative wrapper below.

Definition 3.20 (Conservative sequential learner)**.** We use the same conservative update policy as Yoshinaka's Algorithm 1 [14, Section 4.1]. Let H_0 be the fixed empty-language hypothesis from the learning model. The learner $\mathcal{A}_{f,h}$ starts from H_0. On a pause symbol it keeps its current grammar. On a positive datum w, it keeps the current grammar if that grammar already generates w; otherwise, with K the positive content seen so far, it replaces the hypothesis by $\mathcal{B}_{f,h}(K)$.

Corollary 3.21 (Gold/TxtEx identification)**.** *For every finite-monoid homomorphism $h : \Sigma^* \to M$ and every $f \geq 1$, the conservative learner $\mathcal{A}_{f,h}$ TxtEx-identifies $\mathcal{C}^{\mathrm{mcf}}_{f,h}$ from positive data.*

Proof. MCFG membership is decidable [30], so the learner is computable. Sample consistency gives the invariant that every current hypothesis contains every positive example seen so far: a rebuild contains the whole current content, and a stage without a rebuild leaves a hypothesis that already contains the new datum. Proposition 3.9 makes every rebuilt hypothesis sound.

Fix a stage after the characteristic sample from Theorem 3.19 has appeared. If the current sound hypothesis is already exact, every later positive datum is generated and no further rebuild occurs. Otherwise choose $w \in L$ missing from the current hypothesis. The text eventually presents w, which forces a rebuild from positive content still containing the characteristic sample; that rebuild is exact by Theorem 3.19. From then on no further change is possible. The learner is generally presentation-order-sensitive and need not be set-driven. □

3.5 Polynomial-Time Updates at Bounded Rule Rank

For fixed $f, r_{\max}, h$, the construction is polynomial in the observed sample, as in Yoshinaka's bounded-rank learner [14, Proposition 4 and Section 4.3]. We encode observed tuples by cut positions and sample-word indices, and templates by ordered variables and terminal factors.

Theorem 3.22 (Polynomial construction under a rule-rank cap)**.** *Fix $f, r_{\max} \geq 1$ and a finite homomorphism h. From a finite sample K the complete encoding of $\widehat{G}_{\leq r_{\max}}(K)$ is computable in time $(1 + \|K\|_+)^{O(f r_{\max})}$. Its fan-out is at most f and rule rank at most $r_{\max}$.*

Proof. With $n_K = \max\{1, \|K\|_+\}$, cut positions enumerate all tuple occurrences in $n_K^{O(f)}$ time; fixed-monoid substring products give their types. Grouping by context and type enumerates Rule (Link). Lemma 3.3 enumerates Rule (Comp) and its encodings in $n_K^{O(f r_{\max})}$ time. The remaining rules have rank at most one, and deduplication leaves a polynomial-size complete grammar encoding. □

Corollary 3.23 (Polynomial-time conservative learning at bounded rank)**.** *For fixed $f, r_{\max}, h$, the policy in Definition 3.20 using $\mathcal{B}_{f,h}^{\leq r_{\max}}$ TxtEx-identifies $\mathcal{C}_{f,h}^{\mathrm{mcf}} \cap \mathsf{MCFL}(f, r_{\max})$ in time polynomial in the accumulated positive-sample size per processed datum.*

Proof. Rank-preserving normalization (Proposition 2.8) and soundness and completeness (Propositions 3.9 and 3.18) provide the characteristic sample needed in the proof of Corollary 3.21. Theorem 3.22 bounds rebuilds. Uniform MCFG membership for bounded fan-out and rank is polynomial jointly in the grammar encoding and input length [14, Proposition 4]; the parser saturates finite unary closures. □

At fan-out one every CFG admits a rank-two presentation, with ε handled separately. Thus $\mathcal{A}_{1,h}^{\leq 2}$ covers $\mathcal{C}_{1,h}^{\mathrm{mcf}}$ without implying polynomial characteristic-data length at rank two.

Relation to Yoshinaka's general framework. At bounded rule rank the present typed class is exactly the substitutable class of a finite-sort signature, and our reconstruction generates the same language as Yoshinaka's substructure-indexed $\mathrm{SUBST}_{\mathrm{P}}$ on each sample. These qualitative conclusions are instances of the general framework [15]; precise statements, context-function correspondence, and proofs appear in Appendix B.

4 Characteristic-Data Bounds

Under a rule-rank cap, Proposition 3.17 already gives polynomially many characteristic words; this section controls their total length.

Condition	Characteristic-data control	Source
arbitrary h, bounded rank	polynomial in $\|G\|$ and refined thickness $\theta_{\widetilde{G}_0}$	Lemma 4.3
rule rank one	polynomial in $\|G\|$ alone	Theorem 4.5
$h \preceq h_\Sigma^{k,\ell}$, bounded rank	$O(\|G\|^2 \bar{\theta}_G)$	Theorem 4.9
window plus any finite group	$O(\|G\|^3 \bar{\theta}_G)$	Theorem D.7

The first row is the unconditional baseline for general h. Full yield-type refinement can make its thickness exponentially larger than the original presentation parameters (Proposition 4.4), so that route alone gives no original-thickness bound. Rank one removes the thickness dependence, whereas prefix–suffix-determined typing gives the main higher-rank transfer. The finite-abelian-group

refinement has its direct zero-sum proof in Appendix C; the nonabelian extension is established by a separate order-homogenization proof in Appendix D. The companion CFG result [25] separately excludes a universal original-thickness polynomial for arbitrary finite typing at fan-out one; the semilattice typing of that example is not determined by any fixed prefix–suffix window. Polynomial update time in the observed sample is independent of these data-length refinements (Theorem 3.22 and Corollary 3.23).

4.1 Thickness and reusable canonical contexts

Rule thickness and Yoshinaka's efficiency criterion. For the quantitative results, presentations are reduced and *good*, as in Yoshinaka's fixed-dimension, fixed-rank good form [14, Section 2.3, after Example 2]. For a rule R of G let $\theta_G(R)$ be the minimum length of a successful sentence derivation using R, put $\theta_G := \max_{R\in P} \theta_G(R)$, and define $t_G := \max_A \min\{\|\vec{w}\|_1 : \vec{w} \in L_A(G)\}$. Thus θ_G is rule thickness and t_G nonterminal thickness. Yoshinaka writes these quantities as τ_G and t_G, respectively, with $\tau_G \le \|G\| t_G$ in his notation [14, Section 4.3].

Lemma 4.1 (Polynomial equivalence of the two thickness parameters)**.** *For every reduced good presentation G, $t_G \le \theta_G$ and $\theta_G \le 2\|G\| \max\{1, t_G\}$.*

Proof. For $t_G \le \theta_G$, choose any rule with left-hand side A and place it in a successful derivation. Nondeletion preserves the tuple produced at that occurrence of A, so its total length is bounded by the final sentence. Minimize for A and maximize over A.

For the converse, fix a rule R with left-hand side A and follow a shortest dependency path from the start symbol to A. The path repeats no nonterminal. Keep the path child open, complete each sibling by a shortest terminal tuple, then apply R and complete its children likewise. Across the distinct rules on this construction, both explicit terminal material and child-variable occurrences are bounded by $\|G\|$, giving a successful sentence of length at most $\|G\| + \|G\| t_G \le 2\|G\| \max\{1, t_G\}$. □

Hence the two thickness parameters are polynomially equivalent on reduced good presentations. Yoshinaka's efficiency criterion asks for polynomial hypothesis construction, characteristic-set cardinality, and total positive data in target size and rule thickness [14, Definition 1]; see also [10]. Here characteristic-data bounds concern the capped set-driven reconstruction, while TxtEx stabilization is supplied by the conservative learner. Proposition A.1 transfers the size–thickness scale for reduced λ-free inputs; genuine empty components may instead cause the exponential growth of Proposition A.2.

Lemma 4.2 (Reusable canonical-context bound)**.** *Let $\mathcal{G}_{\mathrm{t}}$ be a trimmed good typed MCFG with n_{typ} non-start typed nonterminals and maximum non-start rule rank at most $r_0 \ge 1$. Suppose that every non-start typed nonterminal Y has a canonical tuple $\omega(Y) \in L_Y(\mathcal{G}_{\mathrm{t}})$ with $\|\omega(Y)\|_1 \le b_{\mathrm{can}}$, and that every non-start typed rule writes at most b_{rule} terminal symbols explicitly. Then the canonical contexts may be chosen so that*

$$|\chi(X)|_{\mathrm{fix}} \le n_{\mathrm{typ}}(b_{\mathrm{rule}} + (r_0 - 1)b_{\mathrm{can}})$$

for every non-start typed nonterminal X. With these contexts and canonical tuples, every selected witness word has length at most

$$n_{\mathrm{typ}}(b_{\mathrm{rule}} + (r_0 - 1)b_{\mathrm{can}}) + b_{\mathrm{rule}} + r_0 b_{\mathrm{can}}.$$

Proof. If $\widetilde{S} \to X$ is present take $\chi(X) = \square_1$; otherwise follow a shortest typed-nonterminal dependency path to X. It has at most n_{typ} non-start steps. Keeping the path child open and filling

each sibling canonically costs at most $b_{\mathrm{rule}} + (r_0 - 1)b_{\mathrm{can}}$ terminals per step. Goodness preserves the selected components, their order, and nonempty separators, giving the context bound. Adding either one canonical tuple or one rule template with at most r_0 child tuples gives the displayed witness bound. □

4.2 Refined-Thickness Baseline

For a reduced good presentation G of an ε-free target, let $\widetilde{G}_0$ be its trimmed full yield-type refinement and put $\bar{\theta}_{\mathrm{typ}} := \max\{1, \theta_{\widetilde{G}_0}\}$, where $\theta_{\widetilde{G}_0}$ is rule thickness computed in the refined grammar. No comparison with the original untyped thickness is assumed.

Lemma 4.3 (Refined-thickness baseline)**.** *Fix $f \geq 1$, $r_{\max} \geq 1$, a finite alphabet Σ, and a finite typing $h : \Sigma^* \to M$. Let G be a reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation of an ε-free language in $\mathcal{C}^{\mathrm{mcf}}_{f,h}$, and let $\widetilde{G}_0$ be its trimmed yield-type refinement. Then the target witness set has cardinality polynomial in $\|G\|$ and total positive length polynomial in $\|G\|$ and $\bar{\theta}_{\mathrm{typ}} := \max\{1, \theta_{\widetilde{G}_0}\}$.*

Proof. The trimmed refinement $\widetilde{G}_0$ remains reduced and good. A minimum-yield successful derivation using a typed rule with left-hand side X contains an X-tuple of total length at most $\bar{\theta}_{\mathrm{typ}}$, so use such tuples canonically. If $\widetilde{S} \to X$ exists take $\chi(X) = \square_1$; otherwise remove the chosen X-subtree from the same derivation to obtain a canonical context with fixed material at most $\bar{\theta}_{\mathrm{typ}}$. Each typed rule writes at most that many terminals and has at most $r_{\max}$ canonical children, so every selected witness has length at most $(r_{\max} + 2)\bar{\theta}_{\mathrm{typ}}$. Therefore

$$|\mathsf{W}(\widetilde{G}_0)| \leq |V||M|^f + |P||M|^{f r_{\max}}, \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ \leq (|V||M|^f + |P||M|^{f r_{\max}})(1 + (r_{\max} + 2)\bar{\theta}_{\mathrm{typ}}),$$

which is polynomial for fixed $f, r_{\max}, h$. □

For trivial typing, no nonterminal is split by yield type; consequently the original thickness scale is retained. This recovers Yoshinaka's untyped characteristic-data estimate [14, Lemma 7]. Without this uniqueness, the refinement may have much larger thickness, as the next proposition demonstrates.

Proposition 4.4 (Exponential blow-up of full yield-type refinement)**.** *There are a fixed two-element typing h_{c}, reduced good fan-out-one rank-two grammars G_n of size $O(n)$, and a constant bound $\theta_{G_n} \leq 2$, such that $L(G_n) = \{\mathsf{a}, \mathsf{c}\}^+$ and the trimmed full yield-type refinement $\widetilde{G}_{n,0}$ satisfies $\theta_{\widetilde{G}_{n,0}} \geq 2^n$.*

Proof. Let $M_{\mathsf{c}} = \{1, \zeta\}$ with $\zeta^2 = \zeta$, $h_{\mathsf{c}}(\mathsf{a}) = 1$, and $h_{\mathsf{c}}(\mathsf{c}) = \zeta$. For $n \geq 1$, view

$$S \to SS \mid \mathsf{a} \mid \mathsf{c} \mid A_0 \mid \cdots \mid A_n,$$
$$A_0 \to \mathsf{a}, \qquad A_i \to A_{i-1}A_{i-1} \mid \mathsf{c} \quad (1 \leq i \leq n)$$

as a fan-out-one MCFG. It is reduced and good, has size $O(n)$ and rule rank two, generates $\{\mathsf{a}, \mathsf{c}\}^+$, and every rule has a successful use of length at most two, so $\theta_{G_n} \leq 2$.

In the full refinement, the retained copy $(A_n)_1$ can derive only words without c; recursively the unique such yield is a^{2^n}. Hence the typed rule $S_{(1)} \to (A_n)_1$ has successful thickness at least 2^n. □

Thus full yield-type refinement admits no bound polynomial in $\|G\| \max\{1, \theta_G\}$ for arbitrary fixed typing, already at fan-out one and rank two. This is a limitation of the full-refinement route, not of the language $\{\mathsf{a}, \mathsf{c}\}^+$ itself.

Prior lower bound at fan-out one. The publicly accessible revised companion CFG manuscript (see the fixed-commit source in [25]) establishes an operator-independent exponential characteristic-data lower bound for fixed two-element semilattice typing, already at fan-out one and rule rank two [25]. We do not reproduce this prior CFG result. It does not automatically extend to higher fan-out, whose tuple substitutability tests more arities.

4.3 Rule Rank One: Grammar-Size Polynomial Data

Unary derivation chains admit a grammar-size bound independent of thickness, even under arbitrary fixed componentwise typing.

Theorem 4.5 (Rank-one polynomial construction and characteristic data)**.** *Fix $f \geq 1$, finite Σ, and a typing $h : \Sigma^* \to M$ into a finite monoid. For any finite standard fan-out-$\leq f$, rank-≤ 1 grammar G_{in} of $L \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$, $\mathcal{B}^{\leq 1}_{f,h}$ is polynomial-time in $\|K\|_+$ and has a characteristic sample of total length $\mathrm{poly}_{f,h}(\|G_{\mathrm{in}}\|)$. The direct-start datum ε is added when $\varepsilon \in L$ without affecting this bound. The conservative learner $\mathcal{A}^{\leq 1}_{f,h}$ consequently TxtEx-identifies $\mathcal{C}^{\mathrm{mcf}}_{f,h} \cap \mathsf{MCFL}(f, 1)$ with polynomial time per datum.*

Proof. Normalize without increasing rank (Proposition 2.8), and trim the typed refinement. Write $n_{\mathrm{typ}} \leq |V||M|^f$ for its number of non-start typed nonterminals and $b_G \leq \max\{1, \|G\|\}$ for the largest explicit terminal count of a rule, taking $b_G \geq 1$. Choose a typed terminal unary derivation that minimizes first total yield length and then the number of rules. No typed nonterminal can repeat: splicing in the later suffix would not increase yield length and would strictly shorten the chain. Therefore $\|\omega(X)\|_1 \leq n_{\mathrm{typ}} b_G$. Lemma 4.2 bounds each witness word by $(2n_{\mathrm{typ}} + 1)b_G$, while Proposition 3.17 gives at most $n_{\mathrm{typ}} + |P||M|^f$ witnesses. Hence

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ \leq (n_{\mathrm{typ}} + |P||M|^f)(1 + (2n_{\mathrm{typ}} + 1)b_G).$$

The polynomial overhead of normalization and the time and convergence bounds follow from Theorem 3.22 and Corollary 3.23. □

This extends the rank-one untyped scale [14, Lemma 7] and the fan-out-one argument of [25] to arbitrary fixed finite typing at every fixed fan-out. Rank two cannot generally yield a grammar-size-only bound: the size-$O(n)$ grammar $A_0 \to \mathsf{c}$, $A_i \to A_{i-1}A_{i-1}$, $S \to A_n$ generates the singleton $\{\mathsf{c}^{2^n}\}$, which is (f, h)-tuple-substitutable for every fixed f, h: over a one-letter alphabet, sharing a sentence context forces equal total component lengths, so every context accepts both tuples or neither. Every characteristic sample for this singleton target contains a word of length at least 2^n.

4.4 Prefix–Suffix-Determined Typings: Thickness-Sensitive Data

Lemma 4.3 gives an arbitrary-h characteristic-data bound in terms of the thickness of the refined typed presentation. Proposition 4.4 shows that this refined thickness need not be polynomially controlled by the original good presentation. A stronger transfer is nevertheless possible whenever the finite typing is determined by a fixed prefix–suffix window. We formulate this boundary-determined regime in the same thickness-sensitive representation model used by Yoshinaka [14, Section 4.3].

Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG and put $\bar{\theta}_G := \max\{1, \theta_G\}$. Here θ_G is the rule-thickness parameter defined above. The argument is stated for the λ-free good core. If the target also contains ε, apply the argument to $L \setminus \{\varepsilon\}$ and add the single datum ε at the start symbol, exactly as in Theorem 3.19; this adds only one length-zero example and does not affect either polynomial bound.

Lemma 4.6 (Rule thickness bounds local terminal material)**.** *Let G be reduced and good. Then:*

(i) for every rule R, the total number of terminal symbols written explicitly in the template of R is at most θ_G;

(ii) for every nonterminal A, there exists a terminal tuple $\vec{u} \in L_A(G)$ with $\|\vec{u}\|_1 \le \theta_G$.

Proof. Reducedness places every rule in some successful derivation. Because a good grammar is nondeleting, every terminal written by that rule and every component of the tuple produced at its left-hand side survives exactly once to the final sentence. A minimum-yield successful derivation using the rule therefore bounds both its explicit terminal material and the total length of that occurrence's left-hand-side tuple. Maximizing over rules gives (i) and (ii). □

Lemma 4.7 (Untyped rule thickness controls canonical contexts)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG and $\widetilde{G}_0$ its trimmed yield-type refinement for any fixed finite typing h. For each $A \in V$ of fan-out d there is an admissible sentence context $E_A \in \mathcal{E}_d$ such that*

$$|E_A|_{\mathrm{fix}} \le \bar{\theta}_G, \qquad E_A[\vec{u}] \in L(G) \quad \text{for all } \vec{u} \in L_A(G),$$

and $E_S = \square_1$ may be chosen. Thus for every non-start typed nonterminal $X = A_{\mathbf{t}}$ the choice $\chi(X) := E_A$ meets Definition 3.14. With canonical tuples satisfying $\|\omega(X)\|_1 \le b_{\mathrm{can}}$ and maximum rule rank $r_{\max}$, every word in $\mathsf{W}(\widetilde{G}_0)$ has length at most $2\bar{\theta}_G + r_{\max} b_{\mathrm{can}}$.

Proof. Take $E_S = \square_1$. For $A \neq S$ choose a source rule R with left-hand side A. Because G is reduced, there is a successful sentence derivation using R and having length $\theta_G(R) \le \theta_G$. Remove the subtree rooted at that occurrence of A, replacing its d components by ordered holes. Lemma 3.7, iterated along the path to the root, and goodness show that the resulting E_A belongs to $\mathcal{E}_d$. Its fixed terminal material is part of the selected sentence, hence has length at most $\theta_G(R)$. Substituting any tuple from $L_A(G)$ gives another successful source derivation, so E_A accepts every such tuple.

By Proposition 3.12(i), $L_X \subseteq L_A(G)$; consequently E_A accepts every tuple of the typed X. The surrounding source derivation can also be lifted to the full typed refinement by Proposition 3.12(ii): the type of X determines all ancestral types given the fixed sibling derivations, and the fresh start symbol admits every start type. In particular, $\chi(X)$ has the required common-context property. For a direct start link, $X = S_{\mathbf{t}}$ and the chosen context is $\square_1$.

A witness is $\chi(X)[\omega(X)]$, $\chi(X)[\vec{c}]$, or $\chi(X)[\rho(\omega(Y_1), \dots, \omega(Y_r))]$. Its context contributes at most $\bar{\theta}_G$ terminals; the source rule writes at most $\bar{\theta}_G$ by Lemma 4.6(i), and it has at most $r_{\max}$ child tuples. The displayed bound follows, with the rank-zero case included. □

Fix $k, \ell \ge 0$ and let $h : \Sigma^* \to M$ be any finite typing with $h \preceq h_\Sigma^{k,\ell}$. Thus h factors through the surjective prefix–suffix typing $h_\Sigma^{k,\ell}$. Let G have fan-out at most f and rule rank at most a fixed $r_{\max} \ge 1$, and let $\widetilde{G}_0$ be its trimmed full yield-type refinement under h. Let n_{typ} be the number of retained non-start h-typed nonterminals.

Lemma 4.8 (Short prefix–suffix-typed canonical tuples)**.** *For every non-start typed nonterminal X of $\widetilde{G}_0$, the canonical tuple may be chosen so that $\|\omega(X)\|_1 \le f(k+\ell+1) n_{\mathrm{typ}} (r_{\max}+1) \bar{\theta}_G$.*

Proof. Choose, among all terminal derivations from X in the trimmed typed grammar $\widetilde{G}_0$, one with the minimum number of grammar-rule nodes, and then mark the root terminals needed to determine its componentwise $h_\Sigma^{k,\ell}$-type: all terminals of a component shorter than $m_0 = \max\{1, k+\ell\}$ from Definition 2.6, and otherwise its first k and last ℓ terminals (one arbitrary terminal when $k = \ell = 0$). There are at most $f(k+\ell+1)$ marked terminals.

No typed nonterminal can occur twice on a root-to-leaf grammar path of this minimal derivation. If an upper and a proper lower occurrence had the same typed label, the lower subtree would itself be a valid typed terminal derivation from the same typed nonterminal, and replacing the upper subtree by it would strictly reduce the number of grammar-rule nodes. Every typed nonterminal occurring in this derivation is retained in $\widetilde{G}_0$; hence every root-to-marked-terminal path contains at most n_{typ} grammar nodes. Their union therefore contains at most $f(k+\ell+1)n_{\mathrm{typ}}$ grammar nodes.

Keep the union of these marked root-to-leaf paths as the retained subderivation. By linearity and nondeletion, each component of every omitted child contributes a contiguous interval to exactly one root component; it is never split between root components. Replace every omitted child subtree by a shortest terminal tuple of its underlying untyped nonterminal. Lemma 4.6(ii) bounds each such tuple by $\bar{\theta}_G$, and part (i) bounds the explicit terminal material of each retained rule by $\bar{\theta}_G$. Because the grammar is good, every replacement component is nonempty. An omitted child cannot contribute to a short root component: every terminal of such a component was marked, so any nonempty contribution from that child would contain a marked terminal and the corresponding root-to-terminal path would place the child on the retained skeleton. In a long root component no replaced block crosses a protected prefix or suffix position. Hence every replacement lies wholly in the unmarked interior of that component: the marked first k and last ℓ terminals remain literal prefixes and suffixes, and the component stays on the long-word side of the threshold. Thus the root prefix–suffix type is unchanged (and when $k=\ell=0$ the single marked terminal preserves nonemptiness). Applying the factor map supplied by $h \preceq h_\Sigma^{k,\ell}$ shows that its componentwise h-type is unchanged as well. Each retained node contributes at most $\bar{\theta}_G$ explicit terminals and has at most $r_{\max}$ omitted child subtrees. The resulting root tuple has length at most $f(k+\ell+1)n_{\mathrm{typ}}(r_{\max}+1)\bar{\theta}_G$. It therefore has the h-type indexing X. The replacements may change internal typed labels of the original typed derivation; this is harmless. We retain only the underlying G-derivation and the preserved root type, and then Proposition 3.12(ii) uniquely lifts that derivation back to G^h with root X. Since X belongs to $\widetilde{G}_0$, the lifted derivation can be inserted into a successful derivation through X and hence survives trimming. Its root tuple may therefore be chosen as $\omega(X)$. □

Apply Lemma 4.7 with $b_{\mathrm{can}} = f(k+\ell+1)n_{\mathrm{typ}}(r_{\max}+1)\bar{\theta}_G$. Every selected witness word then has length at most $2\bar{\theta}_G + r_{\max}b_{\mathrm{can}} = O_{f,r_{\max},k,\ell,h,\Sigma}(\|G\|\bar{\theta}_G)$, since $n_{\mathrm{typ}} \le |V||M|^f$.

Theorem 4.9 (Polynomial data for prefix–suffix-determined typings)**.** *Fix $f \ge 1$, $r_{\max} \ge 1$, k, ℓ, a finite alphabet Σ, and a finite typing $h : \Sigma^* \to M$ satisfying $h \preceq h_\Sigma^{k,\ell}$. Let G be a reduced good fan-out-$\le f$, rank-$\le r_{\max}$ presentation of a language in $\mathcal{C}^{\mathrm{mcf}}_{f,h}$. Then $\mathcal{B}^{\le r_{\max}}_{f,h}$ is polynomial-time in the positive sample. A characteristic sample can be chosen with*

$$|\mathsf{W}(\widetilde{G}_0)| = O_{f,r_{\max},h}(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},k,\ell,h,\Sigma}(\|G\|^2\bar{\theta}_G),$$

where $\bar{\theta}_G = \max\{1,\theta_G\}$. For a reduced λ-free input presentation G_{in} (deleting rules allowed), the structural good-form conversion of Proposition A.1 yields total data polynomial in $\|G_{\mathrm{in}}\|\max\{1,t_{G_{\mathrm{in}}}\}$.

Proof. The h-refinement has at most $|V||M|^f$ typed nonterminals and $|P||M|^{fr_{\max}}$ typed rule instances. Lemma 4.8 applies because preserving the finer $h_\Sigma^{k,\ell}$ boundary type preserves the coarser h-type. Lemma 4.7 now bounds every selected witness word by $O_{f,r_{\max},k,\ell,h,\Sigma}(\|G\|\bar{\theta}_G)$. Proposition 3.17 gives $|\mathsf{W}(\widetilde{G}_0)| \le n_{\mathrm{typ}} + |P||M|^{fr_{\max}}$, with $n_{\mathrm{typ}} \le |V||M|^f$. Multiplying the polynomial number of witnesses by the preceding word-length bound, and adding at most one datum for ε, proves the asserted total-positive-length bound. The same successful witnesses also imply a polynomial bound on the refined typed rule thickness, but that intermediate parameter is not needed for the estimate. Theorem 3.22 gives the hypothesis-construction claim. □

The input-presentation clause follows from Proposition A.1; nonterminal thickness is used because deleting source rules may erase material before good-form conversion. Corollary 3.23 then gives polynomial time per processed datum for the corresponding capped conservative learner.

At fan-out one, this polynomial boundary-typing bound is not new: Yoshinaka's learner for (k, ℓ)-substitutable context-free languages already yields polynomial characteristic data in grammar size and thickness [12, Theorem 1], and the revised companion CFG manuscript extends the window result to finite-group refinements [25]. The contribution of Lemma 4.8 is to preserve boundary information simultaneously across *multiple* tuple components, giving the same kind of bound at every fixed fan-out $f \geq 2$.

The bound is relative to a reduced good bounded-rank presentation. Theorem 5.2 below shows that this higher-rank regime is needed at the language level: it gives a non-context-free $(1, 1)$-typed target of minimum fan-out two and minimum rule rank two even when competing presentations may use arbitrary finite fan-out. Of the examples below, L_{intr} falls under the window theorem, whereas the rank-one languages L_{cd} and $L_\star$ fall under Theorem 4.5; thus each separation is compatible with a quantitative learning bound.

4.5 An Algebraic Criterion for Polynomial Characteristic Data

Write **LI** for the pseudovariety of finite locally trivial semigroups and **Grp** for the pseudovariety of finite groups. Their join $\mathbf{LI} \vee \mathbf{Grp}$ consists of divisors of finite products $S \times A$, with $S \in \mathbf{LI}$ and A a finite group [28, 29]. The following criterion makes the window–group hypothesis intrinsic to the positive image of the fixed typing. Its finite-semigroup ingredient also occurs in the companion CFG manuscript [25].

Lemma 4.10 (Locally trivial–group typings admit window–group refinements)**.** *Let Σ be a nonempty finite alphabet and $h : \Sigma^* \to M$ a finite-monoid morphism. If $h(\Sigma^+) \in \mathbf{LI} \vee \mathbf{Grp}$, then there exist $k, \ell \geq 0$, a finite group A, and a morphism $g : \Sigma^* \to A$ such that*

$$h \preceq (h_\Sigma^{k,\ell}, g) \quad \text{on all of } \Sigma^*.$$

If $h(\Sigma^+) \in \mathbf{LI}$, the group factor can be taken trivial, so $h \preceq h_\Sigma^{k,\ell}$.

Proof. Choose a finite locally trivial semigroup S, a finite group A, a subsemigroup $T \subseteq S \times A$, and an onto semigroup morphism $\pi : T \to h(\Sigma^+)$. For each $a \in \Sigma$ lift $h(a)$ to a pair $(s_a, g_a) \in T$. Multiplying these pairs extends the lifts to a morphism $\rho : \Sigma^+ \to T$ with $\pi\rho = h|_{\Sigma^+}$. Let $s : \Sigma^+ \to S$ and $g : \Sigma^* \to A$ denote the two coordinate morphisms, with $g(\varepsilon) = 1_A$.

The positive image $S_0 = s(\Sigma^+)$ is locally trivial. Put $q = |S_0| + 1$. The standard finite locally trivial identities $S_0^q = S_0 E(S_0) S_0$ and $euf = ef$ for idempotents $e, f \in S_0$ and $u \in S_0$ [25, 27] imply that $s(w)$ is determined by its length-q prefix and suffix whenever $|w| \geq 2q$: writing $w = pxz$ with $|p| = |z| = q$, choose $s(p) = aeb$ and $s(z) = cfd$; writing $s^1(\varepsilon) = 1$ in the adjoined identity monoid, we obtain

$$s(pxz) = ae\,(b\,s^1(x)\,c)\,fd = aefd = s(pz).$$

The parenthesized factor belongs to S_0 even if $x = \varepsilon$, because $b, c \in S_0$. Words shorter than $2q$ are recorded individually by $h_\Sigma^{q,q}$. Therefore equality of $h_\Sigma^{q,q}$-values on Σ^+ implies equality of s-values. Equality of both window and g values then implies equality of ρ values and hence of h values.

The window typing distinguishes ε from every nonempty word, so the kernel inclusion extends to all of Σ^*. Taking $k = \ell = q$ proves the first assertion. If $h(\Sigma^+)$ itself is locally trivial, use $S = h(\Sigma^+)$, its identity projection for π, and the trivial group. □

Corollary 4.11 (Algebraic sufficient conditions for polynomial data)**.** *Fix a nonempty finite Σ, a fan-out cap $f \geq 1$, a rule-rank cap $r_{\max} \geq 1$, and a finite typing $h : \Sigma^* \to M$. Let G be any reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation of a target in $\mathcal{C}^{\mathrm{mcf}}_{f,h}$, and put $\bar{\theta}_G = \max\{1, \theta_G\}$. The capped reconstruction operator has a characteristic sample of cardinality $O_{f,r_{\max},h}(\|G\|)$ with:*

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},h,\Sigma}(\|G\|^2 \bar{\theta}_G), \quad \text{if } h(\Sigma^+) \in \mathbf{LI},$$
$$\|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},h,\Sigma}(\|G\|^3 \bar{\theta}_G), \quad \text{if } h(\Sigma^+) \in \mathbf{LI} \vee \mathbf{Grp}.$$

Proof. For **LI**, apply Lemma 4.10 with trivial group and Theorem 4.9. For $\mathbf{LI} \vee \mathbf{Grp}$, use the finite window–group refinement from that lemma and Theorem D.7. The parameters of the refining morphisms depend only on the fixed typing and alphabet, not on G. The bound on cardinality holds in both cases. □

Remark 4.12. These are *sufficient* algebraic conditions, not a characterization of all typings admitting polynomial original-thickness data. The lower-bound question beyond fan-out one in Open Problem 6.1 remains open. The empty-alphabet case has only the word ε and is trivial.

4.6 Computing Short Characteristic Data

The bounds above are existential: each proof exhibits a short tuple or context. The shortest ones can be computed directly. Call a sentence context of a typed nonterminal X *derivational* if it arises, as in the proof of Lemma 3.13, from a successful typed derivation by deleting one occurrence of the subtree rooted at X. Every such context is admissible for X. The contexts E_A of Lemma 4.7, lifted to $X = A_{\mathbf{t}}$, and those of Lemma 4.2 are derivational.

Proposition 4.13 (Effective short characteristic data)**.** *Fix $f, r_{\max}$ and a finite typing $h : \Sigma^* \to M$. Given a reduced good MCFG G of fan-out at most f and rule rank at most $r_{\max}$, the following can be computed in time polynomial in $\|G\|$, with lengths written in binary:*

(i) $\theta_G(R)$ for every rule R, and hence θ_G and t_G;

(ii) the trimmed refinement $\widetilde{G}_0$ and, for each non-start typed nonterminal X, a canonical tuple $\omega(X) \in L_X$ of minimum total length and a derivational context $\chi(X)$ of minimum fixed length, with $\chi(X) = \square_1$ whenever $\widetilde{S} \to X$ belongs to $\widetilde{G}_0$;

(iii) derivation DAGs of size polynomial in $\|G\|$ representing these tuples and contexts, and hence every word of the witness set $\mathsf{W}(\widetilde{G}_0)$ of Definition 3.15 built from them.

Every word-length and total-length bound proved in this section and in Appendices C and D holds for this computed witness set. Its words can be written out explicitly in time polynomial in $\|G\| + \|\mathsf{W}(\widetilde{G}_0)\|_+$.

Proof. For fixed $f, r_{\max}, h$ the full refinement G^h has at most $|V||M|^f$ typed nonterminals and $|P||M|^{fr_{\max}}$ typed rules, and trimming is a linear-time productivity and reachability computation. View the typed rules as weighted deduction steps. A rule $X \to \rho(Y_1, \ldots, Y_r)$ maps child lengths x_j to $c_\rho + \sum_j x_j$, where c_ρ is the number of terminals written by ρ; this function is superior. To break zero-length ties without cyclic predecessor pointers, minimize the lexicographic pair (terminal length, derivation-node count), adding $(c_\rho, 1)$ at each rule. The pair-valued operation is strictly superior to each child even when no terminal is written. The shortest-derivation algorithm in the weighted deductive form of Nederhof [36], building on Knuth's generalization of Dijkstra's method for the grammar problem (see also Ramalingam and Reps [35]), therefore computes the minimum

length $\lambda(X) = \min\{\|\vec{u}\|_1 : \vec{u} \in L_X\}$ of every typed nonterminal together with an acyclic choice of minimizing rules. These choices form a derivation DAG for $\omega(X)$, with shared subderivations. A second Dijkstra pass computes outside lengths, breaking equal-length ties by the number of edges. It starts with $\mu(X) = 0$ when $\widetilde{S} \to X$ is present. For every typed rule $X \to \rho(Y_1, \ldots, Y_r)$ and every j it adds an edge from X to Y_j of weight $c_\rho + \sum_{i \neq j} \lambda(Y_i)$. The value $\mu(X)$ is the minimum fixed length of a derivational context of X, and the shortest-path tree supplies its DAG. The same two passes on G itself give $t_G = \max_A \lambda(A)$ and $\theta_G(R) = \mu(A) + c_\rho + \sum_j \lambda(B_j)$ for $R : A \to \rho(B_1, \ldots, B_r)$.

Each proof in this section and in the cited appendices bounds a witness word by bounding the length of its context, its rule template, and its canonical child tuples separately. It does so by exhibiting some tuple in L_X and some derivational context of length within the stated bound. The computed minima are no longer, so every stated bound persists. Expanding a DAG takes time linear in the length of the word it represents. □

The singleton $\{\mathsf{c}^{2^n}\}$ after Theorem 4.5 shows that explicit characteristic words can be exponentially longer than $\|G\|$. In that case the derivation DAGs of Proposition 4.13 remain of polynomial size, so the computed sample has a polynomial-size compressed description.

5 Comparison with Multidimensional Substitutability

Yoshinaka's multidimensional substitutability is the direct positive-data baseline [13, 14]. Under trivial typing the present framework reproduces that semantic class; finite typing refines which observed tuples may be identified while preserving compositional rule witnesses. The MCFG proofs additionally require componentwise type propagation and, for the quantitative boundary theorem, simultaneous control of several tuple components. Yoshinaka's (k, ℓ) hierarchy is a separate CFG result [12]; $h_\Sigma^{k,\ell}$ lifts the same boundary information componentwise. Since every (f, h) condition includes arity one, $\mathcal{C}^{\mathrm{mcf}}_{f,h} \cap \mathrm{CFL} \subseteq \mathcal{C}^{\mathrm{mcf}}_{1,h}$. The examples below separate finite typing from the trivial baseline, prefix–suffix typings, and regular-filter explanations.

5.1 Yoshinaka's Baseline and Prefix–Suffix Specialization

Definition 5.1 (Yoshinaka multidimensional substitutability)**.** Using the sentence-context sets $\mathcal{E}_d$ defined in Section 2, a language L is *fD-substitutable* if, for every $1 \leq d \leq f$, all $E, F \in \mathcal{E}_d$, and all $\vec{x}, \vec{y} \in (\Sigma^+)^d$, $E[\vec{x}], \quad E[\vec{y}], \quad F[\vec{x}] \in L \Longrightarrow F[\vec{y}] \in L$. Write $\mathsf{YSub}(f)$ for this semantic class and $\mathsf{YSL}(f, r) := \mathsf{YSub}(f) \cap \mathsf{MCFL}(f, r)$, where r bounds rule rank. We refer to the displayed four-instance implication as the *substitution implication*. Yoshinaka's original definition quantifies over every arity $d \leq f$, uses nonempty tuple components, and requires nonempty internal separators in d-contexts; footnote 3 separately discusses the stronger alternative that permits empty components and separators [14, Section 3.1 and footnote 3].

Recall the finite monoid morphism $h_\Sigma^{k,\ell}$ from Definition 2.6. In Yoshinaka's word-level definition, $w_1, y_1, w_1', w_2, y_2, w_2' \in \Sigma^*$, $u \in \Sigma^k$, and $v \in \Sigma^\ell$, with $uy_1v, uy_2v \neq \varepsilon$; the condition is

$$w_1uy_1vw_1',\ w_1uy_2vw_1' \in L \quad \Longrightarrow \quad (w_2uy_1vw_2' \in L \iff w_2uy_2vw_2' \in L)$$

[12]. Thus the compared fragments uy_iv are nonempty even though the middle words y_i themselves may be empty. Equality of their $h_\Sigma^{k,\ell}$-values is exactly equality of the boundary information used in this condition, including the exact identity of words shorter than $m_0 = \max\{1, k + \ell\}$.

Specializations of the semantic condition. The trivial typing removes the componentwise type premise, so $\mathcal{C}^{\mathrm{mcf}}_{f,h_{\mathrm{triv}}} = \mathsf{YSub}(f) \cap f\text{-MCFL} \subseteq \mathcal{C}^{\mathrm{mcf}}_{f,h}$. At fan-out one, $(1, h^{k,\ell}_{\Sigma})$-substitutability is precisely Yoshinaka's word-level (k, ℓ) condition [12]: equal window types mean identical short words or common prefixes and suffixes of lengths k and ℓ, respectively. The same correspondence applies componentwise at larger fan-out.

Compatibility with Yoshinaka's efficiency scale. For trivial typing no productive nonterminal has more than one yield type. Yoshinaka's untyped grammar witnesses already satisfy $|K_G| \leq |P|$ and $\|K_G\| \leq |P|\tau_G$ [14, Lemma 7], with polynomial time and data for fixed dimension and rank [14, Theorem 1]. Thus the present quantitative claims concern new typed, multi-component settings rather than re-proving the trivial-typing result.

5.2 An Intrinsically Rank-Two Boundary-Typed Target

The next example makes the distinction between the rank-one and higher-rank quantitative regimes intrinsic to the language. Its rule-rank obstruction is not created by finite typing: the balanced-count factor used below already belongs to Yoshinaka's trivial-typing class. The construction instead combines that obstruction with minimum fan-out two and a separation from the trivial-typing class inside one boundary-typed target.

Let

$$\Sigma_{\mathrm{intr}} := \{\mathtt{a}, \mathtt{b}, \mathtt{l}, \mathtt{r}, \mathtt{p}, \mathtt{q}, \mathtt{c}, \mathtt{d}, \#, \$, \mathtt{u}, \mathtt{v}, \mathtt{w}\}$$

and put

$$P_{\mathrm{intr}} := \{\mathtt{lpc}, \mathtt{lpd}, \mathtt{lqc}, \mathtt{rpc}, \mathtt{rpd}\}.$$

Define

$$\mathrm{MIX}^{+}_{2} := \{x \in \{\mathtt{a}, \mathtt{b}\}^{+} : |x|_{\mathtt{a}} = |x|_{\mathtt{b}}\}, \qquad T^{+}_{3} := \{\mathtt{u}^{n}\mathtt{v}^{n}\mathtt{w}^{n} : n \geq 1\},$$

and

$$L_{\mathrm{intr}} := P_{\mathrm{intr}} \# \mathrm{MIX}^{+}_{2} \$ T^{+}_{3}.$$

To make the role of the witness explicit, let

$$\varphi_{\mathrm{mix}} : \{\mathtt{a}, \mathtt{b}\}^{*} \to \mathbb{Z}, \qquad \varphi_{\mathrm{mix}}(\mathtt{a}) = 1, \quad \varphi_{\mathrm{mix}}(\mathtt{b}) = -1.$$

Then $\mathrm{MIX}^{+}_{2} = \{\mathtt{a}, \mathtt{b}\}^{+} \cap \varphi^{-1}_{\mathrm{mix}}(0)$. The language $\{\mathtt{a}, \mathtt{b}\}^{+}$ is trivially f-tuple-substitutable, so Proposition 2.12(ii) gives $\mathrm{MIX}^{+}_{2} \in \mathsf{YSub}(f)$ for every $f \geq 1$; the rank-two CFG for B displayed below also shows that this ingredient is context-free. Thus the intrinsic rank-two phenomenon used here already occurs inside the trivial-typing baseline. The role of L_{intr} is to combine that known rule-rank obstruction with non-context-freeness, minimum fan-out two, and a separation from $\mathsf{YSub}(f)$ in a single finitely typed target.

Theorem 5.2 (A non-context-free intrinsic-rank-two typed witness)**.** *The language L_{intr} satisfies $L_{\mathrm{intr}} \in \mathsf{MCFL}(2, 2) \setminus \mathrm{CFL}$, and, for every $f \geq 2$,*

$$L_{\mathrm{intr}} \in \mathcal{C}^{\mathrm{mcf}}_{f,h^{1,1}_{\Sigma_{\mathrm{intr}}}} \setminus \mathsf{YSub}(f).$$

Moreover, for every finite $f' \geq 1$, $L_{\mathrm{intr}} \notin \mathsf{MCFL}(f', 1)$. Consequently the minimum fan-out of L_{intr} is exactly two, and its minimum rule rank among all finite-fan-out MCFG presentations is exactly two. In particular, Theorem 4.9 applies to a non-context-free language that cannot be covered by Theorem 4.5 at any finite fan-out.

Proof. First we give a reduced good fan-out-two grammar of maximum rule rank two. Let B generate the nonempty balanced-count language by

$$B \to \mathtt{a}B\mathtt{b}B \mid \mathtt{b}B\mathtt{a}B \mid \mathtt{a}B\mathtt{b} \mid \mathtt{ab}B \mid \mathtt{ab} \\ \mid \mathtt{b}B\mathtt{a} \mid \mathtt{ba}B \mid \mathtt{ba}.$$

The usual first-return induction on $|x|_{\mathtt{a}} - |x|_{\mathtt{b}}$ gives $L_B = \mathrm{MIX}_2^+$.

Let the fan-out-two nonterminal A have rules $A \to (\mathtt{u}, \mathtt{w}) \mid (\mathtt{u}\xi_1^1\mathtt{v}, \mathtt{w}\xi_1^2)(A)$. It derives exactly the tuples $\{(\mathtt{u}^n\mathtt{v}^{n-1}, \mathtt{w}^n) : n \geq 1\}$. For every $\alpha \in P_{\mathrm{intr}}$, add the start rule $S \to \alpha\#\xi_1^1\$\xi_2^1\mathtt{v}\xi_2^2(B, A)$. The resulting grammar generates exactly L_{intr}. It is reduced, λ-free, linear, nondeleting, and nonpermuting. It is also nonmerging: the two components of A are separated by the literal terminal $\mathtt{v}$ in every start rule. Hence it is a good fan-out-two grammar of maximum rule rank two.

The language is not context-free. Intersect it with the regular language $R := \mathtt{lpc}\#\mathtt{ab}\$\,\mathtt{u}^+\mathtt{v}^+\mathtt{w}^+$. Then $L_{\mathrm{intr}} \cap R = \{\mathtt{lpc}\#\mathtt{ab}\$\mathtt{u}^n\mathtt{v}^n\mathtt{w}^n : n \geq 1\}$. A homomorphism erasing the fixed prefix and fixing $\mathtt{u}, \mathtt{v}, \mathtt{w}$ maps this language onto $\{\mathtt{u}^n\mathtt{v}^n\mathtt{w}^n : n \geq 1\}$, which is not context-free. Closure of context-free languages under intersection with regular languages and homomorphism therefore shows that L_{intr} is not context-free. Since fan-out-one MCFGs are exactly context-free grammars in the present syntax, while the displayed grammar has fan-out two, the minimum fan-out is exactly two.

For typed membership, let

$$Q_{\mathrm{intr}} := P_{\mathrm{intr}} \,\#\, \{\mathtt{a}, \mathtt{b}\}^+ \,\$\, \mathtt{u}^+\mathtt{v}^+\mathtt{w}^+.$$

This language is strictly 2-local. One complete set of allowed adjacent pairs, including end markers, is

$$\vdash \mathtt{l},\ \vdash \mathtt{r},\quad \mathtt{lp},\ \mathtt{lq},\ \mathtt{rp},\quad \mathtt{pc},\ \mathtt{pd},\ \mathtt{qc},$$
$$\mathtt{c}\#,\ \mathtt{d}\#,\quad \#\mathtt{a},\ \#\mathtt{b},\quad \mathtt{aa},\ \mathtt{ab},\ \mathtt{ba},\ \mathtt{bb},$$
$$\mathtt{a}\$,\ \mathtt{b}\$,\quad \$\mathtt{u},\quad \mathtt{uu},\ \mathtt{uv},\ \mathtt{vv},\ \mathtt{vw},\ \mathtt{ww},\quad \mathtt{w} \dashv .$$

Define the homomorphism $\Phi_{\mathrm{intr}} : \Sigma_{\mathrm{intr}}^* \to \mathbb{Z}^3$ by

$$\Phi_{\mathrm{intr}}(\mathtt{a}) = (1,0,0),\quad \Phi_{\mathrm{intr}}(\mathtt{b}) = (-1,0,0),$$
$$\Phi_{\mathrm{intr}}(\mathtt{u}) = (0,1,0),\quad \Phi_{\mathrm{intr}}(\mathtt{v}) = (0,-1,1),\quad \Phi_{\mathrm{intr}}(\mathtt{w}) = (0,0,-1),$$

and let every other terminal have image $(0,0,0)$. Then $L_{\mathrm{intr}} = Q_{\mathrm{intr}} \cap \Phi_{\mathrm{intr}}^{-1}(0)$. Indeed, the three coordinates impose respectively

$$|x|_{\mathtt{a}} = |x|_{\mathtt{b}}, \qquad |x|_{\mathtt{u}} = |x|_{\mathtt{v}}, \qquad |x|_{\mathtt{v}} = |x|_{\mathtt{w}}.$$

Lemma 5.4 and Proposition 2.12(ii) show that the displayed intersection is $(f, h^{1,1}_{\Sigma_{\mathrm{intr}}})$-tuple-substitutable for every $f \geq 1$. Together with the fan-out-two grammar above, this gives $L_{\mathrm{intr}} \in \mathcal{C}^{\mathrm{mcf}}_{f,h^{1,1}_{\Sigma_{\mathrm{intr}}}}$ for every $f \geq 2$.

The language is outside Yoshinaka's trivial-typing class already at arity one. Take $x := \mathtt{p}$, $y := \mathtt{q}$, and one-hole contexts $E := \mathtt{l}\Box_1\mathtt{c}\#\mathtt{ab}\$\mathtt{uvw}$, $F := \mathtt{r}\Box_1\mathtt{d}\#\mathtt{ab}\$\mathtt{uvw}$. Then $E[x],\ E[y],\ F[x] \in L_{\mathrm{intr}}$, $F[y] \notin L_{\mathrm{intr}}$. Thus the arity-one substitution implication fails, so $L_{\mathrm{intr}} \notin \mathsf{YSub}(f)$ for every $f \geq 1$. Notice again that the boundary typing distinguishes the offending one-letter factors: $h^{1,1}_{\Sigma_{\mathrm{intr}}}(\mathtt{p}) \neq h^{1,1}_{\Sigma_{\mathrm{intr}}}(\mathtt{q})$.

It remains to prove intrinsic rule rank. Put $\mathrm{MIX}_2 := \{x \in \{\mathtt{a}, \mathtt{b}\}^* : |x|_{\mathtt{a}} = |x|_{\mathtt{b}}\}$. Gallot calls an MCFG *non-branching* when every positive-rank rule has a single child nonterminal; in the present terminology this is exactly rule rank at most one, with no globally fixed fan-out

bound [21, Definition 38]. Since every finite grammar has a finite maximum nonterminal arity, Gallot's non-branching MCFL class is $\bigcup_{f'<\infty} \mathsf{MCFL}(f', 1)$. Gallot's Theorem 29 identifies this class with the finite-index EDT0L languages, and Theorem 31 proves that MIX_2 is not finite-index EDT0L. Definition 38 explicitly permits a child variable to occur at most once, and Theorem 28 confirms that non-permuting non-erasing presentations generate the same non-branching class; thus the use of erasing terminal homomorphisms below stays within the formalism of the cited lower bound. Equivalently, MIX_2 is not generated by any non-branching MCFG; Gallot's Propositions 5–6 give the underlying bounded-k-derivation contradiction [21, Theorem 29, Theorem 31, Propositions 5–6].

Bishop et al. later give a peer-reviewed expansion of the same obstruction in the closely corresponding R-MCFG formalism. Their Definition 6.1 uses *rank* for nonterminal arity (the present fan-out), while *non-branching* still means that a propagating rule has a single child nonterminal. Lemma 6.8 supplies the normal form; Definition 7.1 identifies the same balanced language, and Section 7 proves the obstruction, culminating in the stronger non-EDT0L statement of Theorem C [22, Definition 6.1, Lemma 6.8, Definition 7.1, Section 7, Theorem C]. This is corroborating evidence; Gallot's Theorems 29 and 31 remain the direct implication used here.

Now let π fix $\mathtt{a}, \mathtt{b}$ and erase every other terminal in Σ_{intr}. Then $\pi(L_{\mathrm{intr}}) = \mathrm{MIX}_2^+$. If L_{intr} had a fan-out-f' rule-rank-one presentation, applying π to the terminal material in its rule templates would preserve fan-out and rule rank and would give such a presentation of MIX_2^+. Empty terminal pieces created by erasure are permitted in the standard MCFG formalism. Adding a fresh start symbol with one unary identity link to the old start and one rank-zero rule generating ε would then give a fan-out-f' rule-rank-one presentation of MIX_2, contradicting Gallot's non-branching lower bound. Hence $L_{\mathrm{intr}} \notin \mathsf{MCFL}(f', 1)$ for every finite $f' \geq 1$. Together with the displayed rank-two grammar, this proves the claimed intrinsic rule rank. □

The construction is deliberately modular: the MIX_2^+ factor carries the obstruction to rule rank one, while the T_3^+ factor forces the target outside CFL and hence to minimum fan-out two. Thus the theorem establishes the coexistence of the two lower bounds in a single finitely typed target; the two bounds arise from separate factors rather than from an irreducible interaction between the resources.

The non-branching lower bound itself is prior work. What is specific to the present framework is that a strictly local wrapper and the $(1, 1)$ boundary typing place this target inside the typed class, while the displayed four-instance substitution implication keeps it outside Yoshinaka's trivial-typing baseline. The wrapper retains the prior non-branching obstruction, so the same target also has intrinsic rule rank two across all finite fan-outs.

5.3 Finite Typings beyond the Trivial Baseline

The filtering results of Section 2 give useful sufficient constructions, but finite typing is not merely regular post-filtering of the trivial-typing class. The next linear example gives a stronger separation and, at the same time, a sharp fan-out effect.

Let

$$L_{\mathrm{cd}} := \{\mathtt{a}^n\mathtt{c}\mathtt{b}^n : n \geq 0\} \cup \{\mathtt{a}^n\mathtt{d}\mathtt{b}^{2n} : n \geq 0\}$$

over $\Sigma_{\mathrm{cd}} := \{\mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}\}$. Let $M_{\mathrm{cd}} := \{1, \eta_{\mathtt{c}}, \eta_{\mathtt{d}}, 0\}$, where 1 is the identity, 0 is a zero, and every product of two elements of $\{\eta_{\mathtt{c}}, \eta_{\mathtt{d}}, 0\}$ is 0. Define

$$h_{\mathrm{cd}}(\mathtt{a}) = h_{\mathrm{cd}}(\mathtt{b}) = 1, \qquad h_{\mathrm{cd}}(\mathtt{c}) = \eta_{\mathtt{c}}, \qquad h_{\mathrm{cd}}(\mathtt{d}) = \eta_{\mathtt{d}}.$$

Proposition 5.3 (A fan-out-sensitive typed language)**.** *The language L_{cd} is linear and $L_{\mathrm{cd}} \in \mathcal{C}^{\mathrm{mcf}}_{1,h_{\mathrm{cd}}}$. For every $k, \ell \geq 0$, $L_{\mathrm{cd}} \notin \mathcal{C}^{\mathrm{mcf}}_{1,h^{k,\ell}_{\Sigma_{\mathrm{cd}}}}$. There do not exist a language $L_0 \in \mathsf{YSub}(1)$ and a regular language L_{reg} such that $L_{\mathrm{cd}} = L_0 \cap L_{\mathrm{reg}}$. Moreover, for every finite monoid homomorphism $g : \Sigma^*_{\mathrm{cd}} \to N$, $L_{\mathrm{cd}} \notin \mathcal{C}^{\mathrm{mcf}}_{2,g}$.*

Proof. The linear grammar

$$S \to C \mid D, \qquad C \to \mathtt{c} \mid \mathtt{a}C\mathtt{b}, \qquad D \to \mathtt{d} \mid \mathtt{a}D\mathtt{bb}$$

generates L_{cd}.

For $(1, h_{\mathrm{cd}})$-substitutability, let equal-type nonempty factors x, y share an accepting context. Type 1 contains no center symbol, so the context fixes the branch and side and the count constraint forces $x = y$. For type $\eta_{\mathtt{c}}$, writing $x = \mathtt{a}^{n_1}\mathtt{c}\mathtt{b}^{n_2}$ and $y = \mathtt{a}^{n_1'}\mathtt{c}\mathtt{b}^{n_2'}$, common acceptance gives $n_1 - n_2 = n_1' - n_2'$, the invariant for every other $\mathtt{c}$-context. For type $\eta_{\mathtt{d}}$ the invariant is $2n_1 - n_2 = 2n_1' - n_2'$, while type 0 has empty distribution. Hence $L_{\mathrm{cd}} \in \mathcal{C}^{\mathrm{mcf}}_{1,h_{\mathrm{cd}}}$.

Fix k, ℓ and choose $m \geq \max\{1, k + \ell\}$. The words $x = \mathtt{a}^m\mathtt{c}\mathtt{b}^m$ and $y = \mathtt{a}^m\mathtt{d}\mathtt{b}^{2m}$ have the same $h^{k,\ell}_{\Sigma_{\mathrm{cd}}}$-type and share the empty context, but $\mathtt{a}\Box_1\mathtt{b}$ accepts x and rejects y. Thus every prefix–suffix typing fails at fan-out one.

Suppose $L_{\mathrm{cd}} = L_0 \cap L_{\mathrm{reg}}$ with $L_0 \in \mathsf{YSub}(1)$ and L_{reg} regular. Since $\mathtt{c}, \mathtt{d} \in L_0$, substitutability gives $\mathcal{D}_{L_0}(\mathtt{c}) = \mathcal{D}_{L_0}(\mathtt{d})$; using $\mathtt{adb}^2, \mathtt{acb} \in L_0$ then gives $\mathcal{D}_{L_0}(\mathtt{cb}) = \mathcal{D}_{L_0}(\mathtt{cb}^2)$. Consequently $\mathtt{a}^m\mathtt{c}\mathtt{b}^{m+t} \in L_0 \qquad (m \geq 1,\ t \geq 0)$. For m larger than the state number of a DFA for L_{reg}, pumping the final $\mathtt{b}^m$ block of the accepted target word gives some $t > 0$ with the same word in L_{reg}, contradicting the assumed intersection representation.

Finally, for finite $g : \Sigma^*_{\mathrm{cd}} \to N$, choose $1 \leq i < j$ with $g(\mathtt{a}^i) = g(\mathtt{a}^j), \qquad g(\mathtt{b}^i) = g(\mathtt{b}^j)$. The tuples $(\mathtt{a}^i, \mathtt{b}^i)$ and $(\mathtt{a}^j, \mathtt{b}^j)$ have equal componentwise g-type and are both accepted by $\Box_1\mathtt{c}\Box_2$, whereas $\Box_1\mathtt{d}\Box_2\mathtt{b}^i$ accepts only the first: $\mathtt{a}^j\mathtt{d}\mathtt{b}^{i+j} \notin L_{\mathrm{cd}}$ for $i < j$. Thus arity two fails for every finite g. □

Hence L_{cd} separates finite typing from every prefix–suffix typing and arbitrary regular filtering at fan-out one, while no finite typing repairs the language once arity two is tested.

5.4 Local Boundary Control and Arbitrary Regular Filtering

Following the standard local-testability terminology [26], call $L_{\mathrm{sl}} \subseteq \Sigma^*$ *strictly 2-local* if there is a set $\mathsf{Adj} \subseteq (\Sigma \cup \{\vdash, \dashv\})^2$ such that

$$w \in L_{\mathrm{sl}} \quad \Longleftrightarrow \quad \text{every adjacent pair in } \vdash w \dashv \text{ belongs to } \mathsf{Adj},$$

where $\vdash, \dashv \notin \Sigma$ are fixed boundary markers.

Lemma 5.4 (Strictly 2-local constraints are boundary typed)**.** *If $L_{\mathrm{sl}} \subseteq \Sigma^*$ is strictly 2-local, then for every $f \geq 1$, $L_{\mathrm{sl}} \in \mathcal{C}^{\mathrm{mcf}}_{f,h^{1,1}_{\Sigma}}$.*

Proof. Let equal-$h^{1,1}_{\Sigma}$-type nonempty tuples $\vec{x}, \vec{y}$ share an accepting context E_0. Internal adjacent pairs of each y_i are permitted because they occur in $E_0[\vec{y}]$. If $E[\vec{x}] \in L_{\mathrm{sl}}$, every adjacent pair of $E[\vec{y}]$ is either in fixed material, internal to a y_i, or crosses a component boundary. These are permitted respectively by $E[\vec{x}]$, by $E_0[\vec{y}]$, or by equality of the first/last letters encoded by the boundary type; end-marker pairs are unchanged. Thus $E[\vec{y}] \in L_{\mathrm{sl}}$, and symmetry finishes the proof. □

Arbitrary target-dependent regular filters. For comparison, on a fixed alphabet define

$$\mathcal{F}_f^{\mathrm{arb}} := \{L_0 \cap R : L_0 \in \mathsf{YSub}(f),\ R \text{ regular}\} \cap f\text{-MCFL}.$$

This class contains all regular languages (take $\Sigma^* \in \mathsf{YSub}(f)$) and is therefore not TxtEx-identifiable as a single positive-data class by Gold's obstruction. We use it only as an expressiveness baseline, not as an additional learnable class.

The intrinsic-rank-two witness lies in this larger fragment: $L_{\mathrm{intr}} = \Phi_{\mathrm{intr}}^{-1}(0) \cap Q_{\mathrm{intr}}$. It demonstrates intrinsic rank rather than separation from arbitrary regular filtering. Proposition 5.3 gives that separation at fan-out one; the next theorem establishes it for a finitely typed non-context-free fan-out-two target.

Theorem 5.5 (A fan-out-two typed language beyond arbitrary regular filtering)**.** *Let* $\Sigma_\star := \{\mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}\}$ *and*

$$L_\star := \{\mathtt{a}^n \mathtt{c}^{n+1} \mathtt{b}^n : n \geq 0\} \cup \{\mathtt{a}^n \mathtt{d}^{n+1} \mathtt{b}^{2n} : n \geq 0\}.$$

Let $g_\star : \Sigma_\star^* \to (\mathcal{P}(\{\mathtt{c}, \mathtt{d}\}), \cup)$ *be the homomorphism defined by*

$$g_\star(\mathtt{a}) = g_\star(\mathtt{b}) = \varnothing, \qquad g_\star(\mathtt{c}) = \{\mathtt{c}\}, \qquad g_\star(\mathtt{d}) = \{\mathtt{d}\},$$

and put $h_\star := (h^{1,1}_{\Sigma_\star}, g_\star)$*. Then* $L_\star$ *is non-context-free, has minimum fan-out two and minimum fan-out-two rule rank one, and for every* $f \geq 2$, $L_\star \in \mathcal{C}^{\mathrm{mcf}}_{f,h_\star} \setminus \mathcal{F}_f^{\mathrm{arb}}$*. Equivalently, there are no* $L_0 \in \mathsf{YSub}(f)$ *and regular* R_{reg} *such that* $L_\star = L_0 \cap R_{\mathrm{reg}}$*.*

Proof. A good fan-out-two rank-one grammar is

$$\begin{aligned} A_{\mathtt{c}} &\to (\mathtt{ac}, \mathtt{b}) \mid (\mathtt{a}\xi_1^1 \mathtt{c}, \xi_1^2 \mathtt{b})(A_{\mathtt{c}}), \\ A_{\mathtt{d}} &\to (\mathtt{ad}, \mathtt{bb}) \mid (\mathtt{a}\xi_1^1 \mathtt{d}, \xi_1^2 \mathtt{bb})(A_{\mathtt{d}}), \\ S &\to \mathtt{c} \mid \mathtt{d} \mid (\xi_1^1 \mathtt{c} \xi_1^2)(A_{\mathtt{c}}) \mid (\xi_1^1 \mathtt{d} \xi_1^2)(A_{\mathtt{d}}). \end{aligned}$$

It generates exactly $L_\star$. The intersection with the regular language $\mathtt{a}^*\mathtt{c}^+\mathtt{b}^*$ is $\{\mathtt{a}^n \mathtt{c}^{n+1} \mathtt{b}^n : n \geq 0\}$, which is not context-free. For a short closure proof, introduce a fresh symbol # and the homomorphism fixing $\mathtt{a}, \mathtt{b}, \mathtt{c}$ and sending # to $\mathtt{c}$. The inverse image of this branch, intersected with $\mathtt{a}^*\mathtt{c}^*\#\mathtt{b}^*$, is $\{\mathtt{a}^n \mathtt{c}^n \# \mathtt{b}^n : n \geq 0\}$; erasing # would make $\{\mathtt{a}^n \mathtt{c}^n \mathtt{b}^n : n \geq 0\}$ context-free, a contradiction. Thus the minimum fan-out is two, and infinitude together with the displayed grammar gives minimum fan-out-two rule rank one.

For typed substitutability put

$$Q_\star := \mathtt{a}^*\mathtt{c}^+\mathtt{b}^* \cup \mathtt{a}^*\mathtt{d}^+\mathtt{b}^*, \qquad \Psi(w) := (|w|_{\mathtt{a}}, |w|_{\mathtt{b}}, |w|_{\mathtt{c}}, |w|_{\mathtt{d}}).$$

The strictly 2-local language $Q_\star$ is boundary typed by Lemma 5.4. Let equal-$h_\star$-type tuples $\vec{x}, \vec{y}$ share an accepting context E_0, and set $\Delta := \sum_i \Psi(y_i) - \sum_i \Psi(x_i)$. Their equal $g_\star$-types put $E_0[\vec{x}], E_0[\vec{y}]$ on the same branch, so for some $q \in \mathbb{Z}$, $\Delta = q(1,1,1,0)$ or $\Delta = q(1,2,0,1)$ according as that branch is $\mathtt{c}$ or $\mathtt{d}$.

If $F[\vec{x}] \in L_\star$, boundary typing gives $F[\vec{y}] \in Q_\star$, and equal $g_\star$-types put the two fillings on the same F-branch. When E_0 and F choose the same branch, the displayed count difference preserves its affine relation. If they choose different branches, no tuple component can contain $\mathtt{c}$ or $\mathtt{d}$; hence the two center coordinates of Δ vanish, forcing $q = 0$. In either case $F[\vec{y}] \in L_\star$, and symmetry proves $L_\star \in \mathcal{C}^{\mathrm{mcf}}_{f,h_\star}$.

Finally assume $L_\star = L_0 \cap R_{\mathrm{reg}}$ with $L_0 \in \mathsf{YSub}(f)$ and R_{reg} regular. Since $\mathtt{c}, \mathtt{d} \in L_0$, arity-one substitutability gives $\mathcal{D}_{L_0}(\mathtt{c}) = \mathcal{D}_{L_0}(\mathtt{d})$. Hence, replacing the two occurrences of $\mathtt{d}$ successively,

$\mathtt{ad}^2\mathtt{b}^2 \in L_0$ yields first $\mathtt{acdb}^2 \in L_0$ and then $\mathtt{ac}^2\mathtt{b}^2 \in L_0$; together with $\mathtt{ac}^2\mathtt{b} \in L_0$ gives $\mathcal{D}_{L_0}(\mathtt{c}^2\mathtt{b}) = \mathcal{D}_{L_0}(\mathtt{c}^2\mathtt{b}^2)$. Thus $\mathtt{a}^m\mathtt{c}^{m+1}\mathtt{b}^{m+t} \in L_0$ for all $m \geq 1, t \geq 0$. For m at least the state number of a DFA for R_{reg}, pumping the final $\mathtt{b}^m$ block of the accepted target word gives some $t > 0$ with the same word in R_{reg}, hence in the intersection but not in $L_\star$, a contradiction. □

Thus the single target $L_\star$ rules out representation by a Yoshinaka language followed by any finite-state filter, irrespective of how the filter typing refines $h_\star$.

This rank-one separation complements Theorem 5.2, which gives a boundary-typed non-context-free target of minimum rule rank two across all finite fan-outs.

6 Conclusion

Finite-monoid typing gives a compositional bias for positive-data MCFG learning: every fixed (f, h) admits exact reconstruction and TxtEx identification, with polynomial-time updates under a fixed rule-rank cap. Fan-out is part of the bias: L_{cd} shows that increasing tested arity can destroy substitutability beyond repair by any finite typing. The language $L_\star$ additionally excludes arbitrary target-dependent regular filtering.

For characteristic data, arbitrary fixed typing has a grammar-size polynomial at rule rank one, while prefix–suffix-determined typing has a grammar-size-and-rule-thickness polynomial at arbitrary fixed rank. Theorem 5.2 supplies a boundary-typed non-context-free target of minimum fan-out and rule rank two. Full yield-type refinement and empty-component elimination can nevertheless cause exponential thickness growth; Appendix A records the latter normalization details.

Open Problem 6.1 (Original-thickness bounds beyond fan-out one). Fix $f \geq 2$ and a rule-rank bound $r \geq 2$. Does there exist a fixed finite typing h and a family of reduced good fan-out-$\leq f$, rank-$\leq r$ grammars G_n of targets in $\mathcal{C}^{\mathrm{mcf}}_{f,h}$ for which $\|G_n\| \max\{1, \theta_{G_n}\}$ grows polynomially in n but every characteristic sample for $\mathcal{B}^{\leq r}_{f,h}$ has superpolynomial total positive length? The fan-out-one counterexample of the companion CFG paper [25] does not by itself answer this higher-arity question: membership in $\mathcal{C}^{\mathrm{mcf}}_{1,h}$ need not imply membership in $\mathcal{C}^{\mathrm{mcf}}_{f,h}$ for $f \geq 2$.

The fixed-window–finite-group extension is proved in Appendix D; it applies to arbitrary finite groups, with the sharper abelian constant retained in Appendix C. Whether the exponents two and three in the respective window-only and window–finite-group characteristic-data bounds are optimal is open, as is whether the doubly exponential dependence on f of the nonabelian constant $M_{A,f}$ can be reduced. Appendix E extends this original-thickness control to all typings whose positive image is a local group, the largest class excluding the two-element semilattice. The matching lower bound is known only at fan-out one (Corollary E.9). Corollary 4.11 gives a transparent sufficient condition on the positive-image semigroup in $\mathbf{LI} \vee \mathbf{Grp}$. The broader local-group extension and pseudovariety-level dichotomy are treated separately in Appendix E; higher-fan-out lower bounds remain open.

Other directions include adaptive learning over several fan-out biases, learning useful typings without an a priori codomain bound, characterizing the rank-one subfamily inside the typed classes, understanding the relation with full-multicontext congruential MCFGs, and extending the reconstruction to PMCFG-style copying.

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A Details of the Good Normalization

The following bookkeeping records the fixed-parameter bounds used in Proposition 2.8; semantic correctness of the nondeleting and nonmerging conversions is standard [14, 30, 32].

Nondeleting conversion and component permutations. After trimming, use only the nondeleting step (f3) from the proof of Seki et al.'s Lemma 2.2, in Kanazawa's rank-preserving form [30, proof of Lemma 2.2, item (f3)] [32, Section 2.1]. Retained-component subsets give at most $O_f(1)$ decorated copies per nonterminal and $O_{f,r_{\max}}(1)$ projected copies per rule; deleting an empty child cannot increase rank. For nonpermutation, decorate a fan-out-d nonterminal by $\sigma \in \mathfrak{S}_d$. The parent permutation determines the child permutations, so there are at most $f!$ copies, no child is duplicated, and rank is preserved. Denote the result by G_{nd}.

Removing empty tuple components. For a fan-out-d nonterminal A, let

$$\mathsf{Pat}(A) := \{I \subseteq \{1, \ldots, d\} : I \text{ is the nonempty-component pattern of some } \vec{w} \in L_A(G_{\mathrm{nd}})\}.$$

Finite saturation computes these sets. For realizable child patterns, replace variables from absent components by ε, delete empty parent components, and omit all-empty children. There are at most 2^f patterns per nonterminal and $2^{fr_{\max}}$ choices per rule. Introduce A^I for each nonempty pattern I. Retained variables still occur once and in order, so nondeletion/nonpermutation, fan-out, and rank are preserved and the result is λ-free.

Induction on derivations gives L_{A^I} exactly as the projection of source tuples having pattern I; all-empty omitted children can be restored by terminal derivations realizing their all-empty

component pattern. Since the start symbol has fan-out one, $S^{\{1\}}$ generates precisely the nonempty part of the original language. Projection shortens templates, giving $O_{f,r_{\max}}(\|G_{\mathrm{nd}}\|)$ size.

Nonmerging. Apply Yoshinaka's nonmerging conversion [14, Section 2.3, Lemma 1]. Each decorated nonterminal denotes exactly the projected tuples associated with its retained component pattern, and adjacent-component concatenation preserves λ-freeness, nonpermutation, fan-out, and rule rank. For fixed fan-out and rank Yoshinaka's combined good-form conversion has $O(\|G\|)$ size [14, Section 2.3, after Example 2]; trimming can only decrease all bounds. This proves Proposition 2.8.

Thickness transfer for λ-free presentations

Proposition A.1 (Thickness under good-form conversion)**.** *Fix f and $r_{\max}$. Let G_{in} be a reduced λ-free f-MCFG of maximum rule rank at most $r_{\max}$, possibly with deleting rules, and suppose $L(G_{\mathrm{in}}) \cap \Sigma^+ \neq \emptyset$. The good normalization of Proposition 2.8 yields an equivalent reduced good grammar G with the same fan-out and rule-rank bounds such that*

$$\|G\| = O_{f,r_{\max}}(\|G_{\mathrm{in}}\|), \qquad t_G \leq t_{G_{\mathrm{in}}}, \qquad \theta_G = O_{f,r_{\max}}(\|G_{\mathrm{in}}\| \max\{1, t_{G_{\mathrm{in}}}\}).$$

Proof. For λ-free G_{in}, every derived component is nonempty. The nondeleting conversion therefore needs no later empty-pattern split, and each retained decoration A^I generates exactly the projection of $L_A(G_{\mathrm{in}})$ onto I. Hence its minimum total yield cannot exceed that of A. Component permutations preserve total length, and Yoshinaka's nonmerging copies are exact adjacent-component images [14, Section 2.3, Lemma 1], so $t_G \leq t_{G_{\mathrm{in}}}$. All decorations have fixed-parameter linear overhead, giving $\|G\| = O_{f,r_{\max}}(\|G_{\mathrm{in}}\|)$; then Lemma 4.1 yields the displayed bound on θ_G. □

Thus deleting rules do not by themselves obstruct polynomial thickness transfer from a reduced λ-free presentation; genuine empty components are the problematic step.

Proposition A.2 (Empty-component elimination can blow up thickness)**.** *For every $n \geq 1$ there is a reduced fan-out-two, rule-rank-two MCFG E_n of size $O(n)$ with $t_{E_n} \leq 2$ and $\theta_{E_n} = O(n)$ such that the empty-component elimination stage of Proposition 2.8 contains a retained nonterminal and a retained rule of thickness at least 2^n.*

Proof. Let E_n have fan-out-one S, fan-out-two $A_0, \ldots, A_n$, and

$$\begin{aligned} A_0 &\to (\mathtt{a}, \varepsilon), \\ A_i &\to (\varepsilon, \mathtt{c}) \mid (\xi_1^1 \xi_2^1, \xi_1^2 \xi_2^2)(A_{i-1}, A_{i-1}) \quad (1 \leq i \leq n), \\ S &\to (\xi_1^1 \mathtt{d}\, \xi_1^2)(A_n). \end{aligned}$$

It is reduced, nondeleting, nonpermuting, and nonmerging. Short side derivations give $t_{E_n} \leq 2$, and placing any chosen rule on one branch of the binary spine while completing siblings minimally gives $\theta_{E_n} = O(n)$.

If an A_i-tuple has nonempty first component and empty second component, the rule $(\varepsilon, \mathtt{c})$ cannot occur. Inductively the shortest such tuple is $(\mathtt{a}^{2^i}, \varepsilon)$. Thus pattern $\{1\}$ for A_n survives empty-component elimination, and the projected start rule $S \to \xi_1^1 \mathtt{d}\, (A_n^{\{1\}})$ can succeed only with a child of length at least 2^n. Hence the retained nonterminal and rule have thickness at least 2^n. □

This is the same obstruction exhibited by Proposition 4.4: finite state refinement can force exponentially long shortest witnesses inside one refined state.

B Relation to Yoshinaka's General Framework

The bounded-rank learner is a typed instance of Yoshinaka's substitutable-signature framework [15]. We make the correspondence explicit to distinguish prior qualitative learning results from the total-length bounds of Section 4.

Fix finite Σ, $f, r \geq 1$ and $h : \Sigma^* \to M$, and put $M^+ := h(\Sigma^+)$. The signature $\mathfrak{S}_{f,r,h}$ has a start sort $O_* = \Sigma^*$ and tagged sorts

$$T_{d,\boldsymbol{\mu}} = \{\vec{w} \in (\Sigma^+)^d : h^{(d)}(\vec{w}) = \boldsymbol{\mu}\}, \qquad 1 \leq d \leq f, \quad \boldsymbol{\mu} \in (M^+)^d.$$

Its function names comprise all well-typed good MCFG templates of child rank at most r (including rank zero); promotions $T_{1,\mu} \to O_*$; and a nullary ε constant of sort O_*. No function takes an O_* input. After forgetting the type tags, this is a finite sort refinement, in the sense of [15, Section 8], of the signature of good λ-free MCFG operations on nonempty tuples. That untyped signature is a nonempty-component variant of [15, Example 3], rather than literally the signature of that example; at fan-out one the prefix–suffix case corresponds to [15, Example 4]. Write $\mathcal{S}_O, \mathcal{C}_O$ for the substructures and contexts realizable in some grammar over this signature.

Lemma B.1 (Signature contexts are admissible sentence contexts)**.** *For every non-start sort,*

$$\mathcal{S}_{T_{d,\boldsymbol{\mu}}} = T_{d,\boldsymbol{\mu}}, \qquad \mathcal{C}_{T_{d,\boldsymbol{\mu}}} = \{\vec{x} \mapsto E[\vec{x}] : E \in \mathcal{E}_d\}.$$

The latter equality is between functions *on $T_{d,\boldsymbol{\mu}}$. Also $\mathcal{C}_{O_*}$ contains only the identity.*

Proof. Every element of a non-start sort is a rank-zero good template constant. No operation takes O_* as input. For a context of non-start sort, induct along the path from its hole to the start. Composing a good template with an admissible parent context preserves the order and single occurrence of the distinguished child's variables. Consecutive holes in one output component are separated by a terminal or a nonempty sibling component (nonmerging); holes in different output components are separated by a nonempty parent-context separator. Thus every resulting sentence context belongs to $\mathcal{E}_d$, by repeated application of Lemma 3.7.

Conversely, for any $E = v_0 \Box_1 v_1 \cdots \Box_d v_d \in \mathcal{E}_d$, the unary good template $(v_0 \xi_1^1 v_1 \cdots \xi_1^d v_d)$ with a rank-zero child and start promotion realizes $\vec{x} \mapsto E[\vec{x}]$. Here the internal v_i are nonempty, as required by goodness; the output type is fixed by h. □

Proposition B.2 (Bounded-rank coverage by Yoshinaka's 2015 framework)**.** *The signature $\mathfrak{S}_{f,r,h}$ satisfies Assumptions 1–3 of [15], and $\mathsf{Subst}(\mathfrak{S}_{f,r,h}) = \mathcal{C}^{\mathrm{mcf}}_{f,h} \cap \mathsf{MCFL}(f, r)$. Consequently Yoshinaka's identification theorem applies to the bounded-rank class [15, Theorem 1]: for fixed f, r, h, updates are polynomial in the accumulated sample, and characteristic sets have cardinality at most $|N^\sharp| + |P^\sharp|$ for a trimmed signature presentation [15, Lemma 3]. This does not bound the total* length *of those samples.*

Proof. Every signature grammar is a fan-out-$\leq f$, rank-$\leq r$ MCFG after forgetting tags. Conversely, rank-preserving good normalization (Proposition 2.8), trimmed yield-type refinement (Definition 3.11), start promotions and the separate ε constant present every language in $\mathsf{MCFL}(f, r)$ over this signature. At the start sort substitutability is vacuous; at all other sorts Lemma B.1 identifies the four-instance signature implication with Definition 2.10 for each arity and componentwise type. The trimmed presentation has no useless nonterminals or functions, as required in [15, Section 6]. This proves the equality.

For the effective assumptions, sort/context membership and template evaluation are polynomial-time by the lemma and finite typing; uniform bounded-rank MCFG membership is polynomial [14,

Proposition 4]. All operations have arity at most r. For a sentence of length n, enumerate the start word, each tagged tuple of up to f nonoverlapping nonempty intervals with nonempty internal gaps, and the associated context functions. For each parent tuple occurrence, enumerate the rank-$\leq r$ good templates determined by ordered child-component intervals (at most fr such intervals) and the remaining terminal material, recording the actual *function names*. Add promotions and the ε constant as appropriate. By Lemma 3.2, every realizable function instance occurs. At most $2f(r+1)$ cut positions are needed, so the enumeration is polynomial in n for fixed f, r, h. These are precisely Assumptions 1–3; applying Yoshinaka's Theorem 1 and Lemma 3 gives the stated learning consequences. □

Lemma B.3 (Substructure-indexed and guarded reconstruction)**.** *For the preceding signature and enumerations, Yoshinaka's* $\mathrm{SUBST}_{\mathrm{P}}(K)$ *and* $\mathcal{B}^{\leq r}_{f,h}(K)$ *generate the same language for every finite positive sample* $K \subseteq \Sigma^*$.

Proof. Identify their non-start symbols indexed by the same observed tuples and typed sorts. A $\mathrm{SUBST}_{\mathrm{P}}$ rule with parent s_0 and template value $z = \rho(s_1, \ldots, s_t)$ requires a common $c \in \mathrm{CON}_O(K)$ such that $c \odot s_0, c \odot z \in K$, in the notation of [15]. By Lemma B.1 this is an admissible sentence context. The guarded Rule (Link) sends $[s_0]$ to $[z]$, followed by Rule (Comp), or Rule (Const) for $t = 0$, whose witness is materialized by Lemma 3.2. Conversely, every guarded Link is a $\mathrm{SUBST}_{\mathrm{P}}$ instance of the unary identity template; every Comp and Const is its own witnessed parent instance. Induction on derivations therefore equates the non-start symbol languages.

At the start sort the only context is the identity and no operation accepts a start-sort argument. Each observed start symbol consequently has the same language $\bigcup_{w \in K \cap \Sigma^+} L([(w)]) \cup (\{\varepsilon\} \cap K)$, by the promotions and direct ε constant. This is precisely guarded Rule (Start); for $K = \emptyset$ both languages are empty. □

Characteristic witness correspondence. Yoshinaka's proof uses the context-indexed $\mathrm{SUBST}_{\mathrm{D}}$, whereas its Section 7 discusses $\mathrm{SUBST}_{\mathrm{P}}$ under the stated additional assumption [15]. The sample-level equality here is established directly by Lemma B.3. Choosing its non-start witnesses as $c_X = \chi(X)$ and $s_X = \omega(X)$ gives precisely $\mathsf{W}(\widetilde{G}_0)$ at non-start sorts; promoted start witnesses are among these chosen words, and ε contributes its direct datum when present. This identifies the positive-word witness *pattern* with consistent start conventions, not the syntactic hypothesis grammars.

C Windows combined with finite abelian groups

This appendix extends the polynomial total-length theorem to finite-abelian-group refinements of a prefix–suffix window. The original Theorem 4.9, which does not involve a group, remains unchanged.

Lemma C.1 (Short tuples with prescribed abelian-group values)**.** *Let* $G = (V, \Sigma, P, S, \mathrm{fan})$ *be a reduced good MCFG of fan-out at most* f *and rule rank at most* $r_{\max} \geq 1$. *Let* $g : \Sigma^* \to A$ *be a homomorphism into a fixed finite abelian group. Write* $\mathsf{d}(H)$ *for the maximal length of a zero-sum-free sequence over a finite abelian group* H, *so that* $\mathsf{D}(H) = \mathsf{d}(H) + 1$ *is its Davenport constant. For every* $B \in V$ *and every realizable vector* $\mathbf{s} \in A^{\mathrm{fan}(B)}$, *there is* $\vec{u} \in L_B(G)$ *with* $g^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{s}$ *and* $\|\vec{u}\|_1 \leq \beta_g := (r_{\max}+1)\bar{\theta}_G \max\{1, \mathsf{d}(A^f)\,|V||A|^f\}$.

Proof. Choose a terminal derivation T from B whose root has g-vector $\mathbf{s}$, minimizing first the total yield length and then the number of derivation nodes. Label each node by its nonterminal C and the componentwise g-vector of its subtree yield. There are at most $n_g := |V||A|^f$ possible such labels.

If an ancestor and a proper descendant had the same label, replacing the former subtree by the latter would preserve the root g-vector: every MCFG rule computes the parent's group values from the values of its children. The total yield length cannot increase because the rules are nondeleting, and the number of nodes decreases. This contradicts the choice of T. Thus its depth is at most n_g.

Call a node *heavy* if the total length of its subtree yield exceeds $\bar{\theta}_G$. All ancestors of a heavy node are heavy by nondeletion. Suppose $v_1, \dots, v_m$ are pairwise incomparable heavy nodes and the root has fan-out $d \le f$. For each v_i labelled by C_i, choose a terminal derivation from C_i of total length at most $\bar{\theta}_G$, which exists by Lemma 4.6(ii). Replacing the subtree at v_i by this short derivation strictly decreases the total yield length.

Because the grammar is linear and nondeleting, the output components of mutually incomparable subtrees contribute disjoint terminal blocks to the root components, possibly in different orders. In an abelian group, multiplication is additive and the order of those blocks is immaterial. Therefore there are vectors $\delta_i \in A^d$ such that simultaneous replacement of any subset J of the v_i changes the root g-vector by $\sum_{i \in J} \delta_i$. If $m > \mathsf{d}(A^d)$, the sequence $\delta_1 \cdots \delta_m$ over A^d has a nonempty zero-sum subsequence $(\delta_i)_{i \in J}$, so simultaneous replacement at the nodes v_i, $i \in J$, preserves the root vector while strictly shortening its yield. This contradicts minimality. Every heavy antichain consequently has at most $\mathsf{d}(A^d) \le \mathsf{d}(A^f)$ nodes; the inequality holds because A^d embeds in A^f.

If the root is light, its length is at most $\bar{\theta}_G$. Otherwise the heavy nodes form an ancestor-closed tree. Its leaves are a heavy antichain, and every heavy node belongs to a path to one of these leaves. There are therefore at most $\mathsf{d}(A^f)\, n_g$ heavy nodes. Every explicit terminal factor written by a rule is of total length at most $\bar{\theta}_G$ by Lemma 4.6(i). Every light child of a heavy node contributes at most $\bar{\theta}_G$ additional terminals, and a heavy node has at most $r_{\max}$ children. Counting the heavy nodes and their light children gives the asserted bound. The case $|A| = 1$ is covered by the light-root alternative. □

Remark C.2 (Size of the zero-sum constant). The prefix-sum pigeonhole argument gives $\mathsf{d}(A^f) \le |A|^f - 1$, so Lemma C.1 is never weaker than the bound obtained from consecutive partial sums. It is usually much stronger. Write $A \cong C_{n_1} \oplus \cdots \oplus C_{n_s}$ and $n = \exp(A) \ge 2$. Then

$$f \sum_{i=1}^{s} (n_i - 1) \;\le\; \mathsf{d}(A^f) \;\le\; (n-1) + n \log \frac{|A|^f}{n},$$

where the lower bound is attained by a basis sequence and the upper bound is due to van Emde Boas and Kruyswijk [34, Proposition 5.1.8 and Theorem 5.5.5]. If A is a p-group, the lower bound is an equality [34, Theorem 5.5.9] (Olson's theorem). Hence $\mathsf{d}(A^f) = O(\exp(A)\, f \log |A|)$ grows linearly in the fan-out bound, whereas $|A|^f - 1$ grows exponentially; for $A = C_2$ the antichain bound drops from $2^f - 1$ to f.

The value $\mathsf{d}(A^d)$ is also the limit of the antichain step as stated. That step uses only the fact that the difference vectors δ_i have no nonempty zero-sum subsequence, and a zero-sum-free sequence of length $\mathsf{d}(A^d)$ exists. We do not claim that a grammar realizing such an extremal configuration exists.

Lemma C.3 (Short canonical tuples for window–abelian typing)**.** *Assume the hypotheses of Lemma C.1 and fix $k, \ell \ge 0$ and a finite typing $h : \Sigma^* \to M$ satisfying $h \preceq (h_\Sigma^{k,\ell}, g)$. Let $\widetilde{G}_0$ be the trimmed full h-refinement of G, and let n_{typ} be its number of non-start nonterminals. For each such nonterminal X one may choose a canonical tuple $\omega(X)$ with $\|\omega(X)\|_1 \le f(k+\ell+1) n_{\mathrm{typ}} \, (\bar{\theta}_G + r_{\max} \beta_g)$.*

Proof. Choose, among the typed terminal derivations from X, one with the fewest nodes. A repeated typed nonterminal on a path could be eliminated while preserving the root h-type, so each path

has at most n_{typ} nodes. As in the proof of Lemma 4.8, mark in each root component the positions that determine its $h_\Sigma^{k,\ell}$-type: all positions if the component is shorter than the window threshold, otherwise the first k and last ℓ positions, with one position marked when $k = \ell = 0$. The ancestor paths to all marks form a skeleton with at most $f(k+\ell+1)n_{\mathrm{typ}}$ nodes.

Each child subtree omitted from this skeleton lies wholly inside the unmarked interior of every root component to which it contributes. Record its nonterminal C and its actual componentwise g-value $\mathbf{s}$. Replace it by a derivation from C with the same $\mathbf{s}$ and length at most β_g, as supplied by Lemma C.1. Goodness keeps all replacement components nonempty. No protected boundary position changes, and each long root component remains long, so its window type is unchanged. Moreover, each root component is a concatenation of the explicit terminal material and the child components; each replaced child component retains its g-value, so the root g-vector is also unchanged, without requiring a commutativity argument at this stage.

The product $(h_\Sigma^{k,\ell}, g)$ and hence the coarser h-type of the root are unchanged. Forget the intermediate typed labels and uniquely lift the new underlying G-derivation back to the h-refinement using Proposition 3.12(ii). Because X is retained, the resulting root derivation survives trimming. Each skeleton node writes at most $\bar{\theta}_G$ terminals and has at most $r_{\max}$ omitted children. The displayed length estimate follows. □

Theorem C.4 (Polynomial data for window–abelian typing)**.** *Fix $f, r_{\max}, k, \ell, \Sigma$, a finite abelian group A, a homomorphism $g : \Sigma^* \to A$, and a finite typing $h \preceq (h_\Sigma^{k,\ell}, g)$. For every reduced good fan-out-$\le f$, rank-$\le r_{\max}$ grammar G of a language in $\mathcal{C}^{\mathrm{mcf}}_{f,h}$, the capped reconstruction operator $\mathcal{B}^{\le r_{\max}}_{\widetilde{f},h}$ has a characteristic sample satisfying*

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},k,\ell,h,g,\Sigma}(\|G\|^3 \bar{\theta}_G).$$

Proof. The fixed finite typing gives $n_{\mathrm{typ}} \le |V||M|^f = O(\|G\|)$. Lemma C.1 gives $\beta_g = O(\|G\|\bar{\theta}_G)$. By Lemma C.3, the selected canonical tuple lengths are $O(\|G\|^2\bar{\theta}_G)$. Lemma 4.7 therefore bounds each characteristic witness word by $O(\|G\|^2\bar{\theta}_G)$; Proposition 3.17 bounds their number by $O(\|G\|)$. The ε datum, if any, adds only a constant to the encoded total. □

Remark C.5. Commutativity is used in the zero-sum antichain step of Lemma C.1. At fan-out one, subtrees contribute single intervals, and the companion CFG paper treats arbitrary finite groups. For $f \ge 2$ and nonabelian groups, the pieces from different subtrees can interleave in different root components. The zero-sum argument in this appendix does not cover that case. The separate order-homogenization proof in Appendix D does cover arbitrary finite groups, including nonabelian groups.

D Windows combined with arbitrary finite groups

Here we remove commutativity from the finite-group hypothesis in Appendix C. The learning operator and the window-only bound are unchanged. All groups in this appendix are fixed and finite. The constants below are deliberately very large: the result concerns polynomial dependence on the *original* grammar size and rule thickness, not a practical uniform bound in the order of the group.

Lemma D.1 (A homogeneous block-order lemma)**.** *Fix positive integers e, n with $n \ge 3$. For each $t \in [e]$ and $i \in [n]$, let (t, i) be a distinct point on a line, with a total left-to-right order. Suppose that (i) the order of (t, i) and (u, i) is independent of i; and (ii) for each ordered pair (t, u), including*

$t = u$, the order of (t, i) and (u, j) for $i < j$ is independent of the choice of $i < j$. Then the types at the points have a partition into classes, each occupying a consecutive region of the line. Every class has one direction (increasing or decreasing in i); inside its region the points occur in n successive chunks*, one per rank in that direction. Each chunk consists of exactly one point of each type in the class (in a fixed within-chunk order).*

Proof. By (ii) with $t = u$, each type's points are monotone. Consider distinct types t, u. Write X for $(t, i) < (u, j)$ when $i < j$, and Y for $(u, i) < (t, j)$ when $i < j$; both are uniform by (ii). If both types increase, $X \wedge Y$ puts all points of rank i before all points of rank $j > i$. The cases $X \wedge \neg Y$ and $\neg X \wedge Y$ put all t-points before all u-points, or conversely. The remaining case $\neg X \wedge \neg Y$ would give $(t, 1) < (t, 2) < (u, 1) < (u, 2) < (t, 1)$, a contradiction. If t increases and u decreases, $X \wedge Y$ gives $(t, 2) < (u, 3) < (u, 1) < (t, 2)$, while $\neg X \wedge \neg Y$ gives $(t, 3) < (u, 2) < (t, 1) < (t, 3)$. For $X \wedge \neg Y$, all off-diagonal comparisons put t to the left of u; a contrary equal-rank comparison would give $(t, 1) < (u, 2) < (u, 1) < (t, 1)$, contrary to (i). Thus the types are separated. The case $\neg X \wedge Y$ is symmetric. Reverse indices to cover the two-decreasing case. Thus any two types are either *separated*, or have the same direction and are *interleaved by rank*.

Interleaving is transitive. Indeed, if t interleaves with u and u with v, all three have the same direction. Reverse indices to assume increasing. If t and v were separated with t to the left, we would have $(v, 1) < (u, 2) < (t, 3) < (v, 1)$; if v were to the left of t, we would instead have $(t, 1) < (u, 2) < (v, 3) < (t, 1)$. Hence interleaving, with equality, partitions the types into classes. A type outside a class is separated from each type inside it, and must lie on the same side of all its types: otherwise two interleaved types would themselves be separated. The class regions are therefore consecutive. The rank comparisons give successive chunks. Finally the within-chunk order is independent of rank by (i). □

Lemma D.2 (Simultaneous finite-group interval shortening)**.** *Let A be any finite group. Suppose n indices each contribute at most f disjoint blocks to at most f ordered output components, and each block has a prescribed replacement block. A selected index replaces* all *its blocks simultaneously. The other material is fixed. If the individual block destinations and within-index orders are constant across the indices, and all cross-index block order comparisons are constant for $i < j$, then $n \geq \max\{3, |A|^f\}$ implies that a nonempty consecutive interval of indices can be replaced without changing any output component's value in A.*

Proof. Apply Lemma D.1 separately to the block types going to each output component. Across all output components there are $q \leq f$ consecutive class regions. Consider such a region K, and enumerate its chunks $1, \ldots, n$ from left to right. Include all unchanged material between the blocks in the region. Let P_p be the original group value of its prefix ending after chunk p, and Q_p the value of the same prefix when chunks $1, \ldots, p$ have been replaced; set $P_0 = Q_0 = 1_A$. Write $\kappa_p^K = P_p Q_p^{-1}$.

Replacing just chunks $p + 1, \ldots, p'$ gives the following *ordered* value for the entire region:

$$P_p \, Q_p^{-1} Q_{p'} \, P_{p'}^{-1} P_n.$$

Indeed, $Q_p^{-1} Q_{p'}$ is exactly the replacement value of the segment between the two prefix boundaries, and $P_{p'}^{-1} P_n$ is its unchanged suffix. Consequently, the region retains its original value P_n whenever $\kappa_p^K = \kappa_{p'}^K$. This computation uses inverses, but never commutes factors.

If class K increases with the original index order, its chunks for indices $(a, b]$ occupy $(a, b]$. If it decreases, they occupy $(n - b, n - a]$. For $j = 0, \ldots, n$ define a state in A^q by taking κ_j^K from each increasing class and κ_{n-j}^K from each decreasing class. There are $n + 1 > |A|^q$ such states, so two, at indices $a < b$, coincide. Replacing precisely indices $a + 1, \ldots, b$ then preserves each class

region's value. The rest of every output component is unchanged, so all output group values are preserved. □

Lemma D.3 (Polynomial order homogenization). *Let $N \geq 3$, and define $m_0 = N$ and $m_{s+1} = (m_s - 1)^2 + 1$, so that $m_k \leq N^{2^k}$. Let T be a finite set of point types, each assigned to one of several lines. For every index $i \in [n]$ and $t \in T$ let $p_t(i)$ be a point on the line of t, the points on each line being distinct. Let Π be the set of ordered pairs (t, u), including $t = u$, of types on a common line, and put $k = |\Pi|$. If $n \geq m_k$, there is a set $I \subseteq [n]$ of N indices such that, for every $(t, u) \in \Pi$, the truth value of $p_t(i) < p_u(j)$ is the same for all $i < j$ in I.*

Proof. Treat the pairs one at a time, the pairs (t, t) first. Each step passes from a current index set of size at least $(m-1)^2 + 1$ to a subset of size at least m. Homogeneity already obtained, and monotonicity of each type, persist in subsets.

For (t, t), homogeneity means that p_t is monotone on the subset. The Erdős–Szekeres theorem gives a monotone subsequence of length m.

For $t \neq u$, both p_t and p_u are already monotone. Relabel the current indices $1, \ldots, m'$ in order, with $m' \geq (m-1)^2 + 1$. For $i < j$ put $X(i, j) = [p_t(i) < p_u(j)]$. Each row $X(i, \cdot)$ is monotone in j, and each column $X(\cdot, j)$ is monotone in i. Replacing X by $1 - X$ if necessary, we may assume that the rows go from 0 to 1. This does not change which sets are homogeneous.

If the columns go from 1 to 0, put $J(i) = \min\{j > i : X(i, j) = 1\}$, or $J(i) = \infty$ if there is no such j. Then J is nondecreasing. Set $i_1 = 1$ and $i_{s+1} = J(i_s)$ while this is at most m'. For $a < b$ we have $i_b \geq J(i_a)$, so $X(i_a, i_b) = 1$. If the chain has at least m terms, its first m terms suffice. If it has fewer than m terms, its at most $m - 1$ terms cut $[m']$ into at most $m - 1$ intervals $[i_s, i_{s+1})$ and $[i_L, m']$, and one interval has at least $m'/(m-1) > m - 1$ indices. Inside $[i_s, i_{s+1})$, every $i < j$ satisfies $j < i_{s+1} = J(i_s) \leq J(i)$, so $X(i, j) = 0$. The same holds in the last interval, where $J \equiv \infty$.

If the columns go from 0 to 1, the set of pairs with $X = 1$ is closed upward in both coordinates. Let κ be the least index with $X(\kappa, \kappa + 1) = 1$, or $\kappa = m'$ if there is none. For $\kappa \leq i < j$ we have $X(i, j) \geq X(\kappa, \kappa + 1) = 1$. For $i < j \leq \kappa$ we have $X(i, j) \leq X(j - 1, j) = 0$. One of $[1, \kappa]$ and $[\kappa, m']$ has at least $(m' + 1)/2 \geq m$ indices, because $(m-1)^2 + 1 \geq 2m - 1$ for $m \geq 3$.

After all k pairs the remaining set has at least $m_0 = N$ indices. Finally, $m_{s+1} \leq m_s^2$ gives $m_k \leq N^{2^k}$. □

Lemma D.4 (Uniform shortening of heavy antichains). *Set*

$$N_{A,f} = \max\{3, |A|^f\}, \quad U_f = f^{f+1} f!, \quad M_{A,f} = U_f \, N_{A,f}^{2^{f^2}}.$$

Let G be reduced and good, and let $g : \Sigma^* \to A$ be any homomorphism into a finite group. In a terminal derivation minimizing first total yield length and then node count subject to its prescribed componentwise g-value, every antichain of subtrees of total yield length $> \bar{\theta}_G$ has fewer than $M_{A,f}$ nodes.

Proof. Every child component passes through the linear, nondeleting templates to exactly one contiguous block in a root component. Blocks of incomparable subtrees are disjoint. Fix for each node of a heavy antichain a replacement derivation of the *same untyped* nonterminal whose yield has length at most $\bar{\theta}_G$; such a derivation exists by Lemma 4.6(ii). Its group values may differ from the original values.

An antichain node has at most f blocks. Its *unary pattern* records the number of blocks, their root-component destinations, and their order within any shared root component. There are at most U_f such patterns. An antichain of at least $M_{A,f}$ nodes therefore contains at least $N_{A,f}^{2^{f^2}}$ nodes with

one common unary pattern. Enumerate them in any fixed order. The block types of the common pattern are points of at most f types, placed on the root components as lines. There are at most f^2 ordered pairs of types sharing a root component, including $t = u$. Since $m_k \le N^{2^k} \le N^{2^{f^2}}$ for $k \le f^2$, Lemma D.3 yields $N = N_{A,f}$ indices with constant unary pattern and constant cross-index comparisons between (t, i) and (u, j) for all $i < j$.

Within each root component their blocks meet the hypotheses of Lemma D.1. Lemma D.2 therefore supplies a nonempty interval of these selected indices whose simultaneous replacement preserves the *whole root group vector*. Every replaced subtree was heavy but is replaced by one of length at most $\bar{\theta}_G$, so the resulting valid derivation has strictly smaller root yield length. This contradicts its choice. □

Theorem D.5 (Short tuples with prescribed arbitrary finite-group values)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and maximum rule rank $r_{\max} \ge 1$. Let $g : \Sigma^* \to A$ be a homomorphism into* any *finite group A. For every $B \in V$ and every realizable $\mathbf{s} \in A^{\mathrm{fan}(B)}$ there is $\vec{u} \in L_B(G)$ with $g^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{s}$ and*

$$\|\vec{u}\|_1 \le \beta_g^{\mathrm{grp}} := (r_{\max} + 1)\bar{\theta}_G \max\{1, (M_{A,f} - 1)|V||A|^f\}.$$

In particular $\beta_g^{\mathrm{grp}} = O_{A,f,r_{\max}}(\|G\|\bar{\theta}_G)$.

Proof. Choose a terminal derivation from B having the prescribed root vector, lexicographically minimizing (total length, node count). Label each node by its nonterminal and componentwise group value. An ancestor and a proper descendant cannot have the same label: replacing the ancestor by the descendant would preserve every ancestral group vector (by the ordered evaluation of rule templates), would not increase total length (by nondeletion), and would decrease node count. Hence each root-to-leaf path has at most $n_g = |V||A|^f$ nodes.

Call a node heavy when its subtree's total yield length exceeds $\bar{\theta}_G$. Heavy nodes are ancestor-closed. By Lemma D.4, their leaves number at most $M_{A,f} - 1$; all heavy nodes lie on paths to those leaves, so their number is at most $(M_{A,f} - 1)n_g$. If the root is light, its yield already has length at most $\bar{\theta}_G$. Otherwise, each heavy rule contributes at most $\bar{\theta}_G$ explicit terminal symbols, and each of its at most $r_{\max}$ light child subtrees contributes at most $\bar{\theta}_G$. Counting the heavy nodes and their maximal light children gives the displayed upper bound. □

Lemma D.6 (Short canonical tuples for window–arbitrary-group typing)**.** *Under the hypotheses of Theorem D.5, fix $k, \ell \ge 0$ and a finite typing h satisfying $h \preceq (h_\Sigma^{k,\ell}, g)$. Let $\widetilde{G}_0$ be the trimmed full h-refinement of G, and let n_{typ} denote its number of non-start nonterminals. Every non-start X of $\widetilde{G}_0$ admits a canonical tuple $\omega(X)$ of total length at most*

$$f(k + \ell + 1)n_{\mathrm{typ}}(\bar{\theta}_G + r_{\max}\beta_g^{\mathrm{grp}}).$$

Proof. Choose a terminal derivation from X minimizing the number of nodes. Along a root-to-leaf path, a repeated typed nonterminal could be bypassed, preserving the root h-type and decreasing the number of nodes; thus each path has at most n_{typ} nodes. As in Lemma 4.8, mark the root positions determining the $h_\Sigma^{k,\ell}$-value: all positions in a short component, the first k and last ℓ positions in a long component, and one position when $k = \ell = 0$. The union of ancestor paths to the at most $f(k + \ell + 1)$ marks is a skeleton with at most $f(k + \ell + 1)n_{\mathrm{typ}}$ nodes.

For each child subtree omitted from this skeleton, record its underlying nonterminal B and the componentwise group vector actually realized. Theorem D.5 supplies a replacement derivation from B of length at most β_g^{grp} and the *same group value in each component*. Since G is good, all

replacement components are nonempty. No protected position changes, and each root component that was long remains long; hence the window value is preserved. The ordered evaluation of rule templates preserves every root group value because each replaced child component retains its group value; no commutativity is used. Consequently the product $(h_\Sigma^{k,\ell}, g)$ and the coarser root h-type are unchanged. The underlying G-derivation lifts to the trimmed full refinement with root X by Proposition 3.12(ii).

Each skeleton rule contributes at most $\bar{\theta}_G$ explicit terminals and at most $r_{\max}$ omitted children, each contributing at most β_g^{grp} terminals. The displayed bound follows. □

Theorem D.7 (Polynomial characteristic data for window–finite-group typing)**.** *Fix $f, r_{\max}, k, \ell, \Sigma$, any finite group A, a homomorphism $g : \Sigma^* \to A$, and a fixed finite typing h satisfying $h \preceq (h_\Sigma^{k,\ell}, g)$. For every reduced good fan-out-$\leq$ f, rank-$\leq$ $r_{\max}$ grammar G of a target in $\mathcal{C}_{f,h}^{\mathrm{mcf}}$, the capped reconstruction operator $\mathcal{B}_{f,h}^{\leq r_{\max}}$ has a characteristic sample such that*

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},k,\ell,h,g,\Sigma}(\|G\|^3 \bar{\theta}_G).$$

Proof. By Lemma D.6, canonical tuples in the trimmed refinement have length at most

$$b_{\mathrm{can}} \leq f(k+\ell+1) n_{\mathrm{typ}} (\bar{\theta}_G + r_{\max} \beta_g^{\mathrm{grp}}) = O(\|G\|^2 \bar{\theta}_G),$$

since $n_{\mathrm{typ}} \leq |V||M|^f = O(\|G\|)$ and the typing and rank bounds are fixed. Lemma 4.7 bounds each characteristic word by $2\bar{\theta}_G + r_{\max} b_{\mathrm{can}} = O(\|G\|^2 \bar{\theta}_G)$, and Proposition 3.17 bounds the number of words by $O(\|G\|)$. The optional empty-word witness adds only a constant to the encoded total. □

Remark D.8. This strengthening does not assert an original-thickness polynomial for arbitrary finite *monoid* typings. For fixed f the constant $M_{A,f} = U_f \max\{3, |A|^f\}^{2^{f^2}}$ is polynomial in $|A|$. It replaces the multicolor Ramsey bound $U_f\, c_f^{c_f(N_{A,f}-1)+1}$ with $c_f = 2^{f^2}$ used in earlier versions, which is exponential in $|A|^f$. The dependence on f remains doubly exponential. The abelian argument of Appendix C gives a much smaller constant, linear in f, when A is commutative (Remark C.2).

E Local-group typings and the semilattice boundary

The window–group typings of Appendices C and D have positive images in the pseudovariety join $\mathbf{LI} \vee \mathbf{G}$ of finite locally trivial semigroups and finite groups [25, Lemma 7.4]. Here we extend the polynomial original-thickness bound to the strictly larger pseudovariety $\mathbf{LG}$ of *local groups*. This is the class of finite semigroups S in which every local monoid eSe, with e idempotent, is a group. Classically, $\mathbf{LG} = \mathbf{G} * \mathbf{D}$ [37, 38]. We use only the elementary characterization of Theorem E.4, proved below. $\mathbf{LG}$ is exactly the class of finite semigroups containing no copy of the two-element semilattice $U_1 = \{1, \zeta\}$ ($\zeta^2 = \zeta$). That semilattice is the positive image of the typing behind the exponential lower bound of [25]. At the level of pseudovarieties, the semilattice is therefore the only obstruction (Corollary E.9).

Throughout this appendix Σ is nonempty, $k \geq 1$, and A is a finite group. For a word w with $|w| \geq k+1$ let $\sigma_k(w)$ be the sequence of its factors of length $k+1$, ordered by their starting positions.

Definition E.1 (Local-group typing)**.** Let $g : \Sigma^{k+1} \to A$ be any map. For $|w| \geq 2k$ put

$$\gamma_{k,g}(w) := g(F_1) g(F_2) \cdots g(F_{|w|-k}), \qquad \sigma_k(w) = (F_1, \ldots, F_{|w|-k}),$$

the ordered product in A, and put $\gamma_{k,g}(w) := 1_A$ for $|w| < 2k$. The *local-group typing* is

$$\lambda_{k,g}(w) := \big(h_\Sigma^{k,k}(w), \gamma_{k,g}(w)\big).$$

A finite typing h is *local-group-determined* if $h \preceq \lambda_{k,g}$ for some k, A and g.

Recall from Definition 2.6 that, for $\ell = k$, words of length $< 2k$ are *short* and recorded exactly, while a *long* word w is recorded by $\langle \mathrm{pref}_k(w), \mathrm{suf}_k(w)\rangle$. For $s, p \in \Sigma^k$ write $J_g(s,p) := \prod_{i=1}^{k} g((sp)[i..i+k])$. This is the ordered product over the k factors of length $k+1$ of sp that start in s.

Lemma E.2 (Local-group typings are morphisms)**.** *The kernel of $\lambda_{k,g}$ is a congruence of finite index on Σ^*. Hence $\lambda_{k,g}$ is a homomorphism onto a finite monoid $M_{k,g}$. For long u, v,*

$$\lambda_{k,g}(uv) = \big(\langle \mathrm{pref}_k\, u, \mathrm{suf}_k\, v\rangle,\ \gamma_{k,g}(u)\, J_g(\mathrm{suf}_k\, u, \mathrm{pref}_k\, v)\, \gamma_{k,g}(v)\big).$$

Proof. The window component is a morphism by Definition 2.6. For long u, v, the factors of length $k+1$ of uv are, in order: those of u, then the k factors starting in the last k positions of u, then those of v. The middle factors lie inside $\mathrm{suf}_k(u)\,\mathrm{pref}_k(v)$, which gives the displayed formula.

If u is short and v is long, the factors of uv that start in u lie inside $u\,\mathrm{pref}_k(v)$, and the remaining factors are those of v. Thus $\gamma_{k,g}(uv)$ is determined by u, $\mathrm{pref}_k(v)$ and $\gamma_{k,g}(v)$. The case of long u and short v is symmetric. If both are short, uv is determined exactly.

In every case $\lambda_{k,g}(uv)$ is a function of $\lambda_{k,g}(u)$ and $\lambda_{k,g}(v)$. The image has at most $|\Sigma|^{<2k} + |\Sigma|^{2k}|A|$ elements. □

Lemma E.3 (Block decomposition)**.** *Let $w = x_0\beta_1 x_1 \cdots \beta_q x_q$ be a factorization of a word into designated blocks β_j and fixed words x_j. Let $w' = x_0\beta'_1 x_1 \cdots \beta'_q x_q$, where $h_\Sigma^{k,k}(\beta'_j) = h_\Sigma^{k,k}(\beta_j)$ for every j. Then $h_\Sigma^{k,k}(w') = h_\Sigma^{k,k}(w)$. If w is long, there are $c_0, \ldots, c_q \in A$, depending only on the x_j and the window types of the β_j, such that*

$$\gamma_{k,g}(w) = c_0\, \gamma_{k,g}(\beta_1)\, c_1 \cdots \gamma_{k,g}(\beta_q)\, c_q, \qquad \gamma_{k,g}(w') = c_0\, \gamma_{k,g}(\beta'_1)\, c_1 \cdots \gamma_{k,g}(\beta'_q)\, c_q.$$

Proof. A short block is recorded exactly, so $\beta'_j = \beta_j$; absorb such blocks into the fixed words. Every remaining block is long, and its replacement is long with the same first and last k letters. Hence w' is long exactly when w is, and its first and last k letters are unchanged. This gives the window claim.

Classify the factors F of length $k+1$ of w. If F lies inside a long block β_j, it is one of the factors of β_j. Otherwise F meets each long block it touches only in a prefix of length at most k or a suffix of length at most k. If it met both ends of β_j, then $|\beta_j| \le k-1$, so β_j would be short. Hence every factor of the second kind is spelled by the fixed words and the windows $\mathrm{pref}_k(\beta_j)$, $\mathrm{suf}_k(\beta_j)$. The same factors, in the same order, occur in w'. Ordering all factors by starting position, the factors inside β_j form one consecutive run contributing $\gamma_{k,g}(\beta_j)$. Let c_j be the ordered product of the remaining factors between the runs of β_j and β_{j+1}. □

Theorem E.4 (Characterizations of local-group positive images)**.** *For a finite typing $h : \Sigma^* \to M$ put $S := h(\Sigma^+)$. The following are equivalent.*

(i) *$S \in \mathbf{LG}$; that is, eSe is a group for every idempotent e of S.*

(ii) *S has no subsemigroup isomorphic to U_1.*

(iii) *Every idempotent of S lies in the minimal ideal of S.*

(iv) *h is local-group-determined. One may take $k = |S|$.*

Proof. (i)⇒(ii). Let $\{e, f\} \cong U_1$ with $ef = fe = f \neq e$. Then $f \in eSe$ is an idempotent of a group whose identity is e, so $f = e$, a contradiction.

(ii)⇒(iii). Let K be the minimal ideal of S; it contains an idempotent. For an idempotent e of S choose $s \in K$, and let $x := (ese)^\omega$. This is an idempotent in $K \cap eSe$, so $ex = xe = x$. If $x \neq e$ then $\{e, x\} \cong U_1$. Hence $e = x \in K$.

(iii)⇒(i). The minimal ideal K of a finite semigroup is completely simple [39]. For an idempotent $e \in K$ we have $eSe \subseteq K$, so $eSe = eKe$. In a completely simple semigroup this is the $\mathcal{H}$-class of e, a group.

(iii)⇒(iv). By the Rees theorem [39] write $K = \mathcal{M}(H; I, \Lambda; P)$, with product $(i, a, \lambda)(j, b, \mu) = (i, ap_{\lambda j}b, \mu)$. Put $n := |S|$.

Long products lie in K. The Rees quotient S/K has no idempotent other than 0, since all idempotents of S lie in K. Let $s_1, \dots, s_m \in S$ with $m := |S/K| \leq n$, and consider the prefix products in S/K. If none is 0, two of them coincide, say $p_i = p_j$ with $i < j$. Then $p_i = p_i x = p_i x^\omega = 0$ for $x = s_{i+1} \cdots s_j$, a contradiction. Hence $h(w) \in K$ whenever $|w| \geq n$.

Right action. Fix $i_0 \in I$ and for $\lambda \in \Lambda$ let $e_\lambda := (i_0, p^{-1}_{\lambda i_0}, \lambda)$, an idempotent. For $x = (i, a, \lambda) \in K$ we have $xe_\lambda = x$. Hence, for $s \in S$, $xs = x(e_\lambda s)$. Writing $e_\lambda s = (j, c, \mu) \in K$ gives

$$(i, a, \lambda)\, s = (i,\ a\,\rho(\lambda, s),\ \mu(\lambda, s)), \qquad \rho(\lambda, s) := p_{\lambda j} c,$$

where $\rho(\lambda, s) \in H$ and $\mu(\lambda, s) \in \Lambda$ depend only on (λ, s). Dually, the Λ-coordinate of $h(w)$ for $|w| \geq n$ is that of $h(\mathrm{suf}_n w)$, and the I-coordinate is that of $h(\mathrm{pref}_n w)$.

Window factorization. Let $w = a_1 \cdots a_m$ with $m \geq 2n$, and let λ_j be the Λ-coordinate of $h(a_{j-n+1} \cdots a_j)$ for $j \geq n$. Iterating the right action gives

$$h(w) = \Big(i,\ a \prod_{j=n}^{m-1} \rho(\lambda_j, h(a_{j+1})),\ \lambda_m\Big), \qquad h(a_1 \cdots a_n) = (i, a, \lambda_n).$$

The j-th factor of the product depends only on the factor $a_{j-n+1} \cdots a_{j+1}$ of length $n + 1$. Define $g(F) := \rho(\lambda(h(F[1..n])), h(F[n + 1]))$ for $F \in \Sigma^{n+1}$. Then the product is $\gamma_{n,g}(w)$. So $h(w)$ is determined by $\mathrm{pref}_n w$, $\mathrm{suf}_n w$ and $\gamma_{n,g}(w)$. Words of length $< 2n$ are recorded exactly by $h^{n,n}_\Sigma$, and ε is the only word of its $\lambda_{n,g}$-class. Thus $h \preceq \lambda_{n,g}$ with $A = H$.

(iv)⇒(i). **LG** is closed under homomorphic images, so it suffices to show $T := \lambda_{k,g}(\Sigma^+) \in \mathbf{LG}$. A short element is never idempotent: the square of a short word is either longer and short, or long. Let $e = (\langle p, s\rangle, \gamma_e)$ be an idempotent. By Lemma E.2, $\gamma_e J_g(s, p)\gamma_e = \gamma_e$. For every $x \in T$, exe has window $\langle p, s\rangle$, because e is long on both sides. Hence eTe lies in $H_e := \{(\langle p, s\rangle, \alpha) : \alpha \in A\}$. The product on H_e is $\alpha * \beta = \alpha J_g(s, p)\beta$. The map $\alpha \mapsto \alpha J_g(s, p)$ is an isomorphism from $(H_e, *)$ onto A. Thus eTe is a subsemigroup of a finite group, hence a group. □

Proposition E.5 (Window–group typings are local-group typings, strictly)**.** (i) *If $h \preceq (h^{k,\ell}_\Sigma, g)$ for a homomorphism g into a finite group, then $h \preceq \lambda_{K,g'}$ with $K = \max\{k, \ell, 1\}$ and $g'(F) := g(\text{last letter of } F)$. Hence $\mathbf{LI} \vee \mathbf{G} \subseteq \mathbf{LG}$.*

(ii) *Let $\Sigma = \{\mathtt{a}, \mathtt{b}, \mathtt{c}\}$. Let $h_{\mathtt{ab}} := \lambda_{1,g}$ with $g(\mathtt{ab}) = 1 \in \mathbb{Z}_2$ and $g(F) = 0$ otherwise; it records the parity of the number of occurrences of $\mathtt{ab}$, together with the window. Then $h_{\mathtt{ab}}$ is not window–group-determined. Hence $h_{\mathtt{ab}}(\Sigma^+) \in \mathbf{LG} \setminus (\mathbf{LI} \vee \mathbf{G})$.*

Proof. (i) For $|w| \geq 2K$, $g(w) = g(\mathrm{pref}_K w)\, \gamma_{K,g'}(w)$. Shorter words are recorded exactly. The inclusion of pseudovarieties follows from Theorem E.4 and the companion characterization of $\mathbf{LI} \vee \mathbf{G}$ by window–group typings [25, Lemma 7.4].

(ii) Suppose $h_{\mathtt{ab}} \preceq (h_\Sigma^{k,\ell}, g')$ with $g' : \Sigma^* \to A'$. Let $K = \max\{k, \ell, 1\}$ and let q be an even multiple of the exponent of A'. Put $u = \mathtt{a}^K(\mathtt{abc})^q\mathtt{a}^K$ and $v = \mathtt{a}^K(\mathtt{bac})^q\mathtt{a}^K$. Both are long, with prefix $\mathtt{a}^k$ and suffix $\mathtt{a}^\ell$, and $g'(u) = g'(\mathtt{a})^{2K} = g'(v)$. But u contains q occurrences of $\mathtt{ab}$, while v contains exactly one, at $\mathtt{a}^K\mathtt{b}$. So $h_{\mathtt{ab}}(u) \neq h_{\mathtt{ab}}(v)$, a contradiction. Membership in **LG** follows from Theorem E.4. Non-membership in **LI** $\vee$ **G** follows from the companion characterization. □

Lemma E.6 (Short tuples with prescribed local-group type)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and rule rank at most $r_{\max} \geq 1$, and let $\lambda = \lambda_{k,g}$. Let $b_{\mathrm{w}} := f(2k+1)n_{\mathrm{w}}(r_{\max}+1)\bar{\theta}_G$, where $n_{\mathrm{w}} \leq |V||M_{k,k}|^f$ is the number of non-start typed nonterminals of the trimmed $h_\Sigma^{k,k}$-refinement. Put $n_\lambda := |V||M_{k,g}|^f$, and let $M_{A,f}$ be the constant of Lemma D.4. For every $B \in V$ and every realizable $\mathbf{t} \in M_{k,g}^{\mathrm{fan}(B)}$ there is $\vec{u} \in L_B(G)$ with $\lambda^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{t}$ and*

$$\|\vec{u}\|_1 \leq \beta_\lambda := (r_{\max}+1)\, b_{\mathrm{w}} \max\{1, (M_{A,f}-1)\, n_\lambda\}.$$

If A is abelian, $M_{A,f} - 1$ may be replaced by $\mathtt{d}(A^f)$.

Proof. Choose a terminal derivation T from B with root type $\mathbf{t}$ that minimizes first total yield length and then node count. Label each node by its nonterminal and the λ-vector of its yield. As in Lemma C.1, no label repeats on a path, because λ is a morphism (Lemma E.2) and rule templates evaluate homomorphically. Hence every path has at most n_λ nodes.

Call a node *heavy* if its yield has total length greater than b_{w}; heavy nodes are ancestor-closed. Let v be a heavy node labelled by C, and let $\boldsymbol{\omega}$ be the $h_\Sigma^{k,k}$-vector of its yield. Since G is reduced, T can be inserted into a successful derivation. Hence $C_{\boldsymbol{\omega}}$ is a retained typed nonterminal of the trimmed $h_\Sigma^{k,k}$-refinement, and Lemma 4.8 supplies a derivation from C whose yield has the same window vector and total length at most b_{w}. Fix this replacement for every heavy node.

Consider a heavy antichain $v_1, \ldots, v_m$. Each v_i contributes at most f disjoint blocks to the root components. Replacing the subtrees at any set J of indices yields a valid derivation of strictly smaller total length. By Lemma E.3, applied to each root component, the replacement leaves every root window type unchanged. It also changes each long root component's γ-value only through an ordered product in which each block contributes its own γ-value between fixed constants. This is exactly the setting of Lemma D.2, whose proof uses only this ordered-product structure. The proof of Lemma D.4 therefore applies verbatim, with threshold b_{w} in place of $\bar{\theta}_G$, with the window-preserving replacements fixed above, and with γ-values in place of g-values. It uses unary patterns, Lemma D.3 and Lemma D.2. If $m \geq M_{A,f}$, some nonempty set of replacements preserves the whole root λ-vector, contradicting minimality. For abelian A the zero-sum argument of Lemma C.1 applies to the differences $\gamma(\beta') - \gamma(\beta)$ and gives the bound $\mathtt{d}(A^f)$.

Hence the heavy nodes number at most $(M_{A,f}-1)n_\lambda$. Each heavy node writes at most $\bar{\theta}_G \leq b_{\mathrm{w}}$ terminals and has at most $r_{\max}$ light children, each of yield length at most b_{w}. If the root is light, its length is at most b_{w}. □

Theorem E.7 (Polynomial data for local-group typings)**.** *Fix $f, r_{\max}, \Sigma$ and a finite typing h with $h(\Sigma^+) \in$ **LG**. For every reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ grammar G of a language in $\mathcal{C}_{f,h}^{\mathrm{mcf}}$, the capped operator $\mathcal{B}_{f,h}^{\leq r_{\max}}$ has a characteristic sample with*

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},h,\Sigma}(\|G\|^3\bar{\theta}_G).$$

Proof. By Theorem E.4, $h \preceq \lambda_{k,g}$ for some k, A, g depending only on h. Let $X = B_{\mathbf{s}}$ be a non-start typed nonterminal of the trimmed h-refinement and $\vec{u} \in L_X$. Lemma E.6, applied to the λ-vector of $\vec{u}$, gives $\vec{u}' \in L_B(G)$ with the same λ-vector, hence the same h-vector, and $\|\vec{u}'\|_1 \leq \beta_\lambda$. By

Proposition 3.12(ii) its derivation lifts to root X, and it survives trimming as in Lemma 4.8. So canonical tuples of length at most $\beta_\lambda = O(\|G\|^2\bar{\theta}_G)$ exist. Here we use $b_{\mathrm{w}} = O(\|G\|\bar{\theta}_G)$ and $n_\lambda = O(\|G\|)$. Lemma 4.7 bounds every witness word by $2\bar{\theta}_G + r_{\max}\beta_\lambda$, and Proposition 3.17 bounds their number by $O(\|G\|)$. □

Theorem E.7 contains the window and window–group theorems as special cases (Proposition E.5(i)), with the same exponent three in the group case. The intersection lemma below makes the lower-bound side precise.

Lemma E.8 (Type filters preserve substitutability)**.** *If L is (f,h)-tuple-substitutable and $F \subseteq M$, then $L \cap h^{-1}(F)$ is (f,h)-tuple-substitutable.*

Proof. Let $\vec{x}, \vec{y}$ have equal h-vectors and share a context E in $L' := L \cap h^{-1}(F)$. They share E in L, so they have the same L-distribution. If $E'[\vec{x}] \in L'$, then $E'[\vec{y}] \in L$ and $h(E'[\vec{y}]) = h(E'[\vec{x}]) \in F$, so $E'[\vec{y}] \in L'$. □

Corollary E.9 (Pseudovariety dichotomy)**.** *Let $\mathbf{V}$ be a pseudovariety of finite semigroups.*

(i) *If $\mathbf{V} \subseteq \mathbf{LG}$, then for every alphabet, every typing h with $h(\Sigma^+) \in \mathbf{V}$, and all fixed $f, r_{\max}$, characteristic data of total length $O(\|G\|^3\bar{\theta}_G)$ exist as in Theorem E.7.*

(ii) *If $\mathbf{V} \not\subseteq \mathbf{LG}$, then $U_1 \in \mathbf{V}$. At fan-out one there is a typing h_{U} with $h_{\mathrm{U}}(\Sigma^+) = U_1$ and targets with $\|G\|, \theta_G = O(n)$, every characteristic sample of which for $\mathcal{B}^{\leq 2}_{1,h_{\mathrm{U}}}$ contains a word of length $3 \cdot 2^n - 2$. The same holds for every set-driven operator admitting characteristic samples throughout $\mathcal{C}^{\mathrm{mcf}}_{1,h_{\mathrm{U}}}$.*

Proof. (i) is Theorem E.7. For (ii), choose $S \in \mathbf{V} \setminus \mathbf{LG}$. By Theorem E.4, S has a subsemigroup isomorphic to U_1, and pseudovarieties are closed under subsemigroups.

Take the doubling family L_n, the typing h_{U} and the long type-1 word w^*_n of [25, Proposition 6.3]. There L_n is shown to be Clark–Eyraud substitutable, hence $(1, h_{\mathrm{U}})$-tuple-substitutable. By Lemma E.8, $L^-_n := L_n \cap h_{\mathrm{U}}^{-1}(\zeta) = L_n \setminus \{w^*_n\}$ is as well. If $K \subseteq L_n$ omits w^*_n, then $K \subseteq L^-_n$, and Proposition 3.9 gives $L(\mathcal{B}^{\leq 2}_{1,h_{\mathrm{U}}}(K)) \subseteq L^-_n$. Thus every characteristic sample contains w^*_n. The set-driven statement is the distinguishing-pair argument of [25, Proposition 6.3(iv)] for the pair L_n, L^-_n. □

Remark E.10 (Scope of the dichotomy).

(i) Part (i) holds at every fixed fan-out. Part (ii) is established only at fan-out one. The doubling family does not lift directly: L_2 is not $(2, h_{\mathrm{U}})$-tuple-substitutable. The type-1 pairs $([, \mathtt{a}])$ and $([\mathtt{a},])$ share the context $[[\mathtt{aa}]\,\square_1\,\mathtt{a}\,\square_2\,]$ but have different distributions. This was checked exhaustively for $n \leq 3$. Under the finer typing recording each component's set of letters, which still has semilattice image, the corresponding failure appears at $n = 3$. Whether part (ii) holds at fan-out $f \geq 2$ is the pseudovariety form of Open Problem 6.1.

(ii) The dichotomy concerns pseudovarieties. For an individual typing h with $h(\Sigma^+) \notin \mathbf{LG}$, the class $\mathcal{C}^{\mathrm{mcf}}_{1,h}$ need not contain a coded copy of the doubling family. Whether every such typing forces superpolynomial characteristic data remains open.

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